

**The Impact of Non-Traditional Agricultural Exports
on Sustainable Development in developing
countries: A case study of the Ghanaian economy.**

Diploma Thesis

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Abstract

Owusu Agyei, G. (2026). The Impact of Non-Traditional Agricultural Exports on Sustainable Development in developing countries: A case study of the Ghanaian economy. Diploma thesis, Mendel University in Brno, Brno.

Ghana's economy has long depended on gold, timber, and crude oil, with gold alone generating about USD 13 billion in 2025. Severe environmental damage from illegal mining highlights the need to expand Non-Traditional Agricultural Exports (NTAEs) such as palm olein, processed cocoa products, shea butter, coconuts, cashew, and yams. This thesis analyses the socio-economic and ecological impacts of NTAEs using 2000-2025 data, assessing effects on GDP, employment, and the ecological footprint within Ghana's diversification agenda. Using ARDL, and ECM models, the study shows NTAEs enhance economic resilience but face challenges including environmental degradation, quality-control issues, and exchange-rate volatility.

Keywords

Non-Traditional Agricultural Exports (NTAEs), GDP, Employment, Ecological Footprint, Sustainability, ARDL, ECM, Export-Led Growth.

Abstrakt

Owusu Agyei, G. (2026). Dopad netradičního zemědělského exportu na udržitelný rozvoj v rozvojových zemích: případová studie ghanské ekonomiky. Diplomová práce, Mendelova univerzita v Brně.

Ghanská ekonomika se dlouhodobě opírá o vývoz zlata, dřeva a ropy, přičemž samotné zlato v roce 2025 generovalo přibližně 13 miliard USD. Závažná environmentální poškození způsobená nelegální těžbou však zdůrazňují nutnost rozšířit vývoz netradičních zemědělských produktů (NTAEs), jako jsou palmový olein, zpracované kakaové výrobky, bambucké máslo, kokosové produkty, kešu ořechy a jamy. Tato práce analyzuje socioekonomické a ekologické dopady NTAEs na základě dat z let 2000-2025 a hodnotí jejich vliv na růst HDP, zaměstnanost a ekologickou stopu v rámci ghanské strategie diverzifikace exportu. Pomocí modelů ARDL a ECM studie ukazuje, že NTAEs posilují ekonomickou odolnost, avšak čelí výzvám, jako je degradace životního prostředí, problémy s kontrolou kvality a volatilita směnných kurzů.

Klíčová slova

Netradiční zemědělské exporty (NTAEs), HDP, zaměstnanost, ekologická stopa, udržitelnost, ARDL, ECM, exportně orientovaný růst.

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1 Introduction

1.1 Background of Study

Ghana's export sector remains a major driver of economic development through foreign exchange generation, employment creation, and improved trade balances (Raghutla, 2020). Between 2019 and 2023, total exports amounted to approximately US\$ 116.4 billion, with revenues in 2019 and 2020 exceeding national GDP (OECD, 2025; World Bank, 2023). In 2022, gold dominated exports at US\$ 9.53 billion, followed by crude oil and cocoa beans, but this dependence exposes Ghana to global price volatility, as demonstrated by the sharp commodity downturn of 1980-1983. To mitigate such vulnerabilities, Ghana strengthened its diversification agenda through the Ghana Export Promotion Authority (GEPA) and the Economic Recovery Programme, promoting Non-Traditional Exports (NTEs) a broad category encompassing manufactured goods, handicrafts, processed foods, and agricultural products (Ayambila, 2023).

Within this broader category, non-traditional agricultural exports (NTAEs) represent a distinct subset comprising plant-based commodities such as cashew, shea, coconut, pineapples, yam, and refined palm olein (Abban, 2023). Unlike NTEs generally, NTAEs specifically target high-value agricultural markets and are increasingly promoted as environmentally sustainable alternatives to mineral extraction due to their renewable nature and potential for ecological restoration. Although NTE earnings reached US\$ 3.9 billion in 2023, challenges including quality-control issues, global price fluctuations, and declining agricultural investment continue to constrain NTAE performance, underscoring the need for sustained policy support (GEPA, 2023).

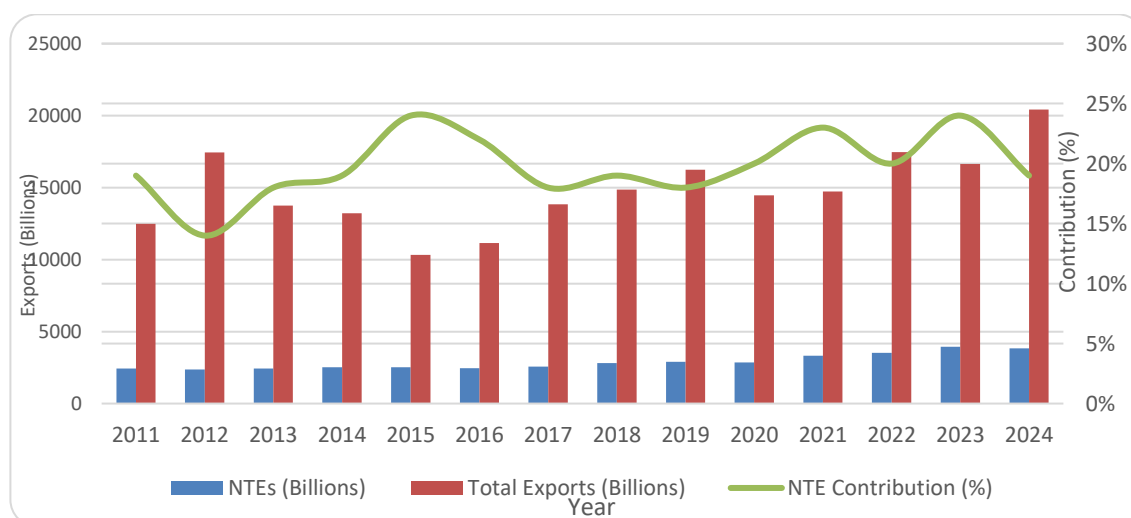


Figure 1 Non-Traditional Exports Performance

Source: Author's Design with data from GEPA 2024 report

Figure 1 highlights the rising importance of Non-Traditional Exports (NTEs) in Ghana's export portfolio. NTE earnings reached US\$3.944 billion in 2023, while their share of national exports increased from 18 % in 2013 to 24 % in 2023, reflecting a 33 % growth in contribution (GEPA, 2023). This expansion aligns with Ghana's diversification strategy aimed at reducing dependence on traditional commodities. In 2024, the Manufactures/Semi-Processed subsector dominated NTEs at 83.44 %, followed by Agricultural exports at 13.48% and Handicrafts at 3.08% (GEPA, 2024). Despite overall progress, Non-Traditional Agricultural Exports (NTAEs) declined by 3.02% in 2023, with earnings falling from US\$511.27 million to US\$494.12 million. GEPA attributes this contraction to low production, price volatility, quality-control challenges, and shifting consumer preferences (GEPA, 2023). Commodity-specific trends varied: cashew and natural rubber declined, while shea oil exports grew by 26.51 %. These patterns underscore both the resilience and vulnerabilities within Ghana's NTE and NTAE sectors, emphasizing the need for strengthened standards and targeted investment.

Table 1 Recent contribution of Sub-Sectors to Total NTEs In 2024 vis-à-vis 2023 (In nominal price)

Sector	2023 (US\$)	2024 (US\$)	Contribution to Total NTEs
<i>Agricultural</i>	494,115,223	515,418,088	13.48
<i>Handicrafts</i>	95,736,322	117,749,229	3.08
<i>Manufacturers & Semi-Processed</i>	3,354,295,172	3,191,382,259	83.44

Source: Author's Design with data from GEPA 2024 report

In 2024, the Manufactures and Semi-Processed subsector dominated Ghana's Non-Traditional Exports (NTEs) at 83.44 %, while Agricultural exports rose to 13.48% and Handicrafts to 3.08% (GEPA, 2024). This dominance highlights the centrality of value addition and industrialization in Ghana's export strategy. Given the rising importance of Non-Traditional Agricultural Exports (NTAEs), assessing the factors influencing their performance and contribution to economic growth remains essential (GEPA, 2024).

1.2 Statement of the Problem

Although NTAEs have expanded by 3,210% since the 1990s (GEPA, 2020), existing publications largely emphasize export volumes while neglecting their socio-economic and sustainability implications (Abban, 2023). Key gaps include limited understanding of sector actors, livelihood impacts, and environmental consequences. Addressing these omissions, this thesis investigates the factors influencing NTAE growth and sustainability to strengthen Ghana's policy and development outcomes.

1.3 Research Questions

In this thesis, the following research questions will be examined:

- How are Ghana's Non-Traditional Agricultural Exports linked to employment outcomes and selected environmental indicators between 2000 and 2025?
- What identifiable factors support or hinder the sustainable growth of Ghana's Non-Traditional Agricultural Exports?
- What policy interventions can promote the sustainable development of Ghana's Non-Traditional Agricultural Exports while minimizing its sustainability effects?

1.4 Research Objectives

The main objective of this thesis is to examine how Ghana's NTAEs such as cashew nuts, refined palm olein, cocoa powder, etc. affect the domestic economic performance, employment dynamics, and the socio-ecological environment.

Whereas existing studies focuses on NTEs, this study focuses on NTAEs with the specific objectives:

- To assess the relationship between Non-Traditional Agricultural Exports of Ghana and the GDP growth of Ghana.
- To ascertain the factors that promote and impede the sustainability of Non-Traditional Agricultural Exports.
- To emphasize the effects of the export of Non-Traditional Agricultural Products on the socio-ecological environment and the economy.
- To make recommendations to the government of Ghana on actions to undertake to grow the positive impacts that exporting non-traditional agricultural product bring.

1.5 Organization of the Study

This study is organized into five (5) main chapters. Chapter one dealt with the introduction of the study which involved the background of the study, the statement of the problem, the objective of the study, the significance of the study, and scope of the study and the limitations of the study. Chapter two would depict an inquiry of important literature of other authors in relation to the topic under study. The literature review explains the basic terminologies of NTAEs and its socio-economic impact on the Ghanaian economy and the socio-ecological environment. Chapter three would thus concentrate on the methodology of the study. Chapter four would deal mainly with the presentation and analysis of collected data. Finally, chapter five shall focus on the summary of findings, conclusions as well as the recommendations.

2 LITERATURE REVIEW

2.1 INTRODUCTION

This chapter is both theoretical and empirical review of the concept of Non-Traditional Agricultural Exports and their macroeconomic, labour-market, and environmental implications for Ghana. The literature available records an exponential rise in NTAE performance, increasing from US 45 million in the year 2000 to over US\$ 710 million in 2025 (GEPA, 2025), however empirical evidence remains fragmented and often obscured by aggregated classifications. This section examines the core concepts of how NTAEs contribute to GDP, employment, and the socio-ecological environment in light of Ghana's vision to achieve the UN SDGs 1, 8, and 12 which has to deal with No Poverty, Decent Work and Economic Growth and Responsible Consumption and Production respectively. By classifying agricultural exports as a single category, a large portion of the literature now in publication obscures the differences between traditional and non-traditional export goods (UNCTAD, 2023; AfDB, 2023).

Drawing on export-led growth theory, structural transformation, global value chain frameworks, and the Environmental Kuznets Curve, the chapter identifies methodological inconsistencies, limited sector-specific datasets, and insufficient causal analysis as key gaps in current research. These insights inform the development of hypotheses and the conceptual framework guiding the thesis's econometric investigation of NTAE-driven structural change.

2.2 Theoretical Foundations

The study is anchored in four (4) main theoretical frameworks: Export-Led Growth (ELG) theory, Structural Transformation, Global Value Chains (GVC), and the Environmental Kuznets Curve (ECK).

2.2.1 Export-Led Growth Theory

While the export-led growth (ELG) hypothesis has long emphasized the role of exports in stimulating economic performance (Feder, 1983; Frankel & Romer, 1999), recent literature stresses that export composition (not just export volume), determines long-term development outcomes (Hausmann et al., 2007; Lee, 2021). This shift in thinking is particularly relevant for Ghana, where traditional exports such as cocoa, gold, and timber have historically dominated the export basket but have shown vulnerability to price shocks and limited domestic value addition (GSS, 2023; World Bank, 2022).

According to the ELG hypothesis, export growth boosts economic growth through productivity spillovers, economies of scale, foreign exchange earnings, and technology transfer. Export-oriented economies typically grow more quickly, according to both more recent cross-country data (World Bank, 2020; IMF, 2021) and classic studies (Feder, 1983; Helpman & Krugman, 1985).

Nevertheless, the structural distinctions across export categories are hidden since these researches frequently depend on aggregate export data.

This issue is demonstrated by Ghana's export basket, where gold and cocoa account for the majority of overall exports yet differ greatly in terms of value addition, employment intensity, and environmental effect (GSS, 2023).

2.2.2 Structural Transformation

The long-term movement of economies from agriculture to industry and services, caused by the reallocation of labour from low-productivity to higher-productivity sectors, is described by structural transformation theory (Lewis, 1954; Kuznets, 1973). Since productivity increases in non-agricultural industries boost total national revenue, this shift is thought to be crucial for long-term economic growth. Contemporary development literature also shows there is a connection between export diversification, advancements in technology, skills, and institutional capacity in addition to sectoral shifts for effective structural transformation (McMillan & Rodrik, 2011). Classical dual-sector models argue that economic development occurs when surplus labour in subsistence agriculture is absorbed into expanding industrial sectors, raising aggregate productivity and incomes. Empirical studies show that countries achieving sustained growth typically experience declining agricultural employment shares alongside rising manufacturing output and labour productivity (Herrendorf et al., 2014). Structural transformation is therefore both a cause and consequence of economic modernization.

For resource-dependent economies such as Ghana, the process is essential for reducing vulnerability to commodity-price shocks and enhancing employment opportunities, particularly for youth and women. Recent evidence highlights that productivity gains arise not only from sectoral shifts but also from within-sector upgrading driven by value addition and improved market access (Diao et al., 2019). Within this framework, non-traditional agricultural exports (NTAEs) are viewed as catalysts for structural change by promoting agro-processing, expanding rural non-farm employment, and strengthening linkages between agriculture and industry. This theoretical foundation informs the empirical analysis undertaken in this thesis.

2.2.3 Global Value Chains (GVC)

GVC literature supports this viewpoint by emphasizing how involvement in globally dispersed production systems can hasten structural transformation.

According to Gereffi, Humphrey, and Sturgeon (2005) and Taglioni and Winkler (2016), integration into GVCs allows nations to specialize in particular production stages, access international markets, and gain from knowledge transfer, standards upgrading, and employment creation across multiple nodes of the value chain. GVC participation can create jobs in farming, processing, logistics, certification, and distribution for agricultural exporters, especially those involved in Non-Traditional Agricultural Exports (NTAEs), extending prospects beyond primary production.

For Ghana's NTAE sector, the GVC lens enables assessment of how value addition, standards compliance, and chain governance influence export performance and long-run structural transformation.

2.2.4 Environmental Kuznets Curve (EKC) and Environmental Implications of Export Growth

According to the Environmental Kuznets Curve (EKC) framework, which looks at the dynamic relationship between economic growth and environmental degradation, environmental pressure rises with income at first but eventually falls after a certain point (Grossman & Krueger, 1995; Stern, 2004). Three (3) mechanisms explain this inverted-U pattern: composition effects, where structural shifts toward less resource-intensive sectors change environmental outcomes; technique effects, where technological advancements and tighter regulations lessen environmental harm; and scale effects, where increased production intensifies resource use and pollution (Panayotou, 1993; Dasgupta et al., 2002).

The EKC offers a helpful lens for comprehending how export-driven growth may concurrently generate environmental concerns and opportunities in the context of agricultural export expansion, especially Non-Traditional Agricultural Exports (NTAEs). Intensification of high-value crop production, for example, can lead to deforestation, soil depletion, or excessive use of chemicals, but integration into international markets can also encourage adherence to sustainability standards, better farming practices, and environmentally conscious certification programs (Lambin & Thorlakson, 2018). This study's conceptualization of NTAEs as a multifaceted driver of development—one that influences employment patterns, economic performance, and environmental results in interrelated ways, is informed by the EKC framework, structural transformation, and global value chain theories.

2.3 Conceptual Framework

This study positions Non-Traditional Agricultural Exports (NTAEs) within an integrated conceptual framework that links economic growth, employment, and environmental sustainability. Grounded in the Export-Led Growth (ELG) hypothesis, Structural Transformation theory, Global Value Chains (GVC), and the Environmental Kuznets Curve (EKC), the framework conceptualizes NTAEs as drivers of GDP through market diversification, productivity spillovers, and foreign-exchange earnings (Grossman & Krueger, 1995; Stern, 2004). Employment effects emerge through value-chain upgrading, labour reallocation, and skill development, while environmental outcomes reflect scale, composition, and technique effects. Feedback loops highlight how environmental degradation can constrain agricultural productivity, whereas economic gains may stimulate further investment in NTAEs.

Despite the recognized role of exports in development, limited research isolates the multidimensional impacts of NTAEs in Ghana (Jambor, 2020). Key gaps include scarce time-series analyses, weak integration of environmental sustainability, and the absence of holistic frameworks. This study addresses these gaps using 2000-2025 data and robust econometric techniques.

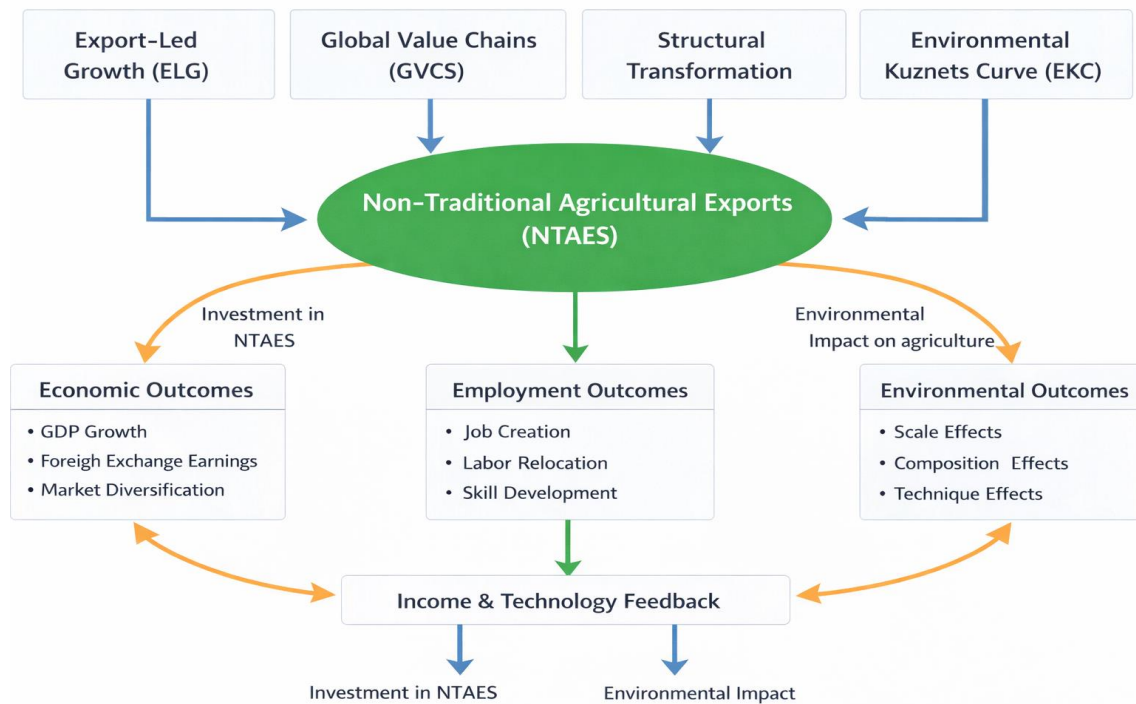


Figure 2 Conceptual Framework Diagram
Source: Author's design with Copilot prompts

2.4 NTAEs Exports and Economic Growth

Agricultural products that do not belong to a nation's historically dominating export commodities are referred to as non-traditional agricultural exports (NTAEs). Horticultural goods, fish and shellfish, prepared foods, handicrafts, and newly developed high-value crops like pineapples, mangoes, cashew, shea, and vegetables are among Ghana's NTAEs (GEPA, 2023). NTAEs are newer or formerly domestically consumed goods that are now produced for international markets, in contrast to conventional exports like cocoa, gold, and lumber (Ampadu-Agyei, 1995; Ayambila, 2023). Their changing characteristics are a reflection of Ghana's larger diversification initiatives, which are meant to boost rural employment, increase export revenue, and adapt to shifting demands in the global market.

2.4.1 Global and Regional Evidence on NTAEs

Non-Traditional Agricultural Exports (NTAEs) are now a major tool for rural development, export diversification, and integration into high-value international markets. High-value horticulture exports, such as grapes, avocados, and berries, have been effectively increased in Latin American nations like Chile and Peru, leading to higher foreign exchange profits, rural employment, and technological advancement (FAO, 2020; OECD, 2021). These nations serve as examples of how focused investments in export standards, cold-chain logistics, and irrigation can boost rural economies' competitiveness in international value chains.

Kenya and Ethiopia are two prominent examples of effective NTAE spread in Sub-Saharan Africa. As one of the top exporters of cut flowers worldwide, Kenya's floriculture sector employs a large number of women and young people and incorporates thousands of smallholders into export supply chains (World Bank, 2020; Minot & Ngigi, 2021). According to UNCTAD (2022) and Alemu & Trienekens (2023), coordinated public-private partnerships, foreign investment, and adherence to international quality standards have all contributed to Ethiopia's rapid increase in vegetable and flower exports. The advantages of NTAEs are not shared equally, notwithstanding these achievements. According to recent studies, smallholders that lack the resources to comply may be marginalized by strict sanitary and phytosanitary (SPS) regulations, certification requirements, and buyer-driven governance structures (Narayanan, 2021; Tröster & Hiete, 2021).

In areas where horticulture is expanding quickly, there is evidence of water depletion, soil deterioration, and pesticide usage, all of which have increased environmental concerns (Barbier & Burgess, 2019; Amoako et al., 2023). These inconsistent results emphasize the necessity of context-specific evaluations, especially in nations like Ghana, where NTAEs are becoming more well-known but function in different institutional, ecological, and commercial environments.

2.4.2 Ghana's Export Structure and the Rise of NTAEs

Cocoa, gold, and crude oil made up more than 70% of all export revenue in recent years, according to national economic figures. Non-Traditional Exports (NTEs), on the other hand, have increased gradually and will reach noteworthy levels by 2024. Non-Traditional Agricultural Exports (NTAEs), a subset of NTEs; include shea butter, cashew nuts, pineapples, and mangoes have grown significantly within NTEs. According to national agricultural performance reports, NTAEs currently make up a sizable portion of all NTE profits. Notwithstanding these advancements, obstacles still exist, such as restricted financial access, poor infrastructure, and adherence to global norms.

Table 2 Composition of Ghana's Exports (Selected Years)

Year	Cocoa (%)	Gold (%)	Timber (%)	Oil (%)	NTAE (%)	Other (%)
2000	32	41	9	0	3	15
2005	30	39	8	0	4	19
2010	28	43	7	0	6	16
2013	21	37	6	23	8	5
2015	20	36	5	25	9	5
2018	19	40	5	22	10	4
2020	21	44	5	18	10	2
2021	20	43	4	19	11	3
2022	19	42	4	20	12	3
2023	19	42	4	20	12	3

Source: Author's design with data from GSS (2023), GEPA (2023), World Bank DWI (2023), BOG Annual Reports & MOTI Trade Statistic

Ghana's intentional diversification strategy through horticulture, cashew, shea, and processed agricultural goods is seen in the rise in NTAEs from 3% of total exports in 2000 to 12% in 2023. This expansion is consistent with GEPA's National Export Development Strategy (NEDS 2020-2029), which emphasizes NTAEs as drivers of value addition and employment in rural areas.

2.4.3 NTAEs and Economic Growth in Ghana

Exports and Ghana's economic growth have been extensively researched. The beneficial effect of exports on GDP is confirmed by both earlier research and more contemporary evaluations. However, the majority of research ignores the unique function of NTAEs by using aggregate export data. In order to close this gap, this study uses time-series data from 2000 to 2025 to examine how NTAEs contribute to GDP growth. Higher value addition (e.g., processed cocoa into powder and paste, cashew, shea butter), market diversification (EU, Middle East, Asia), decreased susceptibility to shocks to commodity prices, and backward links to rural economies are some of the ways that NTAEs support growth.

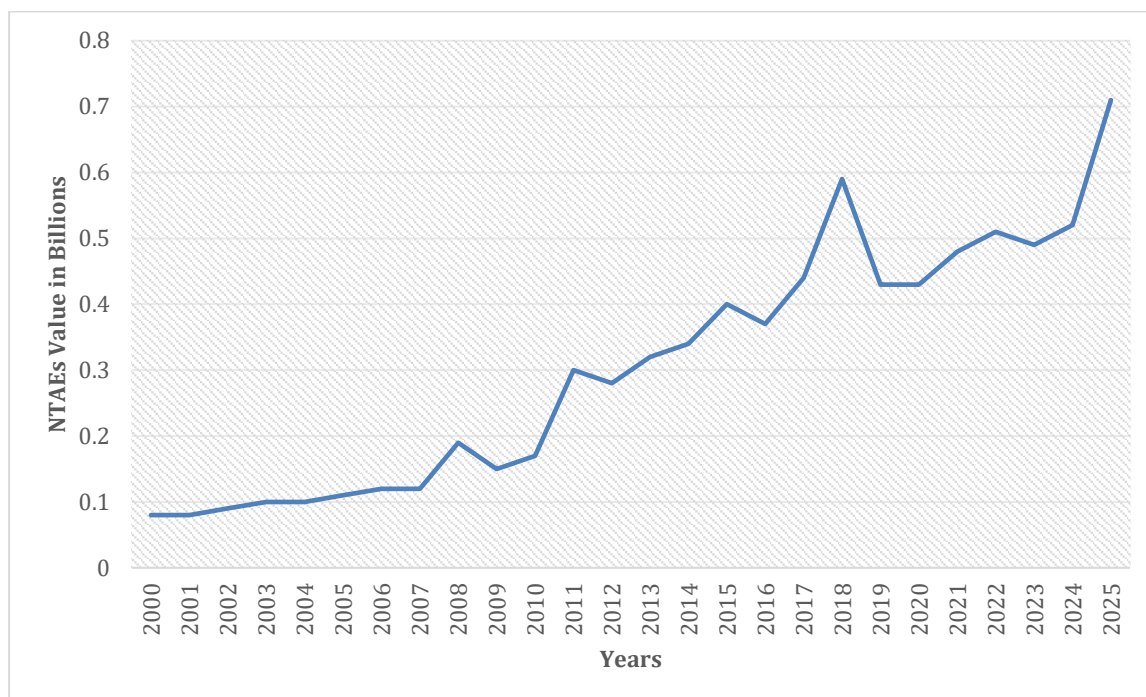


Figure 3 Ghana's NTAE Values in Billions of USD (2000-2025)

Source: Author's own design with data from GEPA (2025)

From the above diagram, Ghana's NTAE trajectory from 2000 to 2025 shows a steady and strong increase tendency, increasing from US\$ 50 million to more than US\$ 700 million. Horticulture, cashew, shea, seafood, and processed agricultural products are the main drivers of this eight-fold increase. The consistent yearly increase, which averages between 8 and 9%, shows how resilient the industry is in contrast to traditional exports, which have seen more fluctuation.

Furthermore, the tendency is consistent with the National Export Development Strategy's strategic interventions by GEPA. Overall, the results highlight NTAEs' increasing significance as a dynamic force behind Ghana's economy's structural reform, rural employment, and export growth. The disaggregated analysis of the contribution of Non-Traditional Agricultural Exports (NTAEs) to Ghana's economy, as presented in this thesis, addresses a critical gap in the existing literature. Current studies largely provide aggregated assessments that obscure sector-specific dynamics; this research therefore seeks to illuminate the distinct economic, employment, and environmental dimensions of NTAEs, thereby advancing empirical understanding of their developmental significance.

2.4.4 NTAES and GDP of Ghana

Gross Domestic Product (GDP) as the total monetary value of all goods and services produced within a country over a specific period, usually annually or quarterly; serves as a key indicator of a country's economic growth and performance (Wong, 2021). Across Africa, agriculture which is the base for NTAEs, accounts for 30-40% of GDP. This chapter will be devoted to evaluating the texts concerning the terms and elements that influence the performance of NTAEs. The measures listed by some authors as ways to increase Ghana's NTAEs' contributions to GDP will also be examined in this section of the thesis.

As per the Food and Agriculture Organization (FAO), countries like Cambodia have an agricultural GDP of 34.2% of its total GDP of about US\$ 6.187 billion in 2005, Bangladesh in that same year, had Agricultural GDP of about 20.1% out of their total GDP of US\$ 60,034 million. Afghanistan recorded 36.1% Agricultural GDP out of its US\$ 7,308 billion total GDP. In recent times (i.e. 2002/2023), the World Bank Group data reflects that, emerging economies in Africa like Nigeria, Rwanda, Kenya, Liberia, Guinea, Guinea Bissau, Ghana, Somalia has Agriculture contributing 22.7%, 27.1%, 21.8%, 35.8%, 29.1%, 30.7%, 21.1%, and 55.8% of its GDP respectively (World Bank Group, 2025).

In 2019, agriculture contributed roughly 20% to Ghana's GDP, underpinning employment and industrial inputs. In 2018, national GDP grew 8.4%, while agricultural output expanded 6.3%, underscoring the sector's macroeconomic significance (Nyamekye, 2021). Because primary commodities like crude oil, raw agricultural products, and unprocessed minerals; make up more than 80.0% of Ghana's overall exports, the country is susceptible to external shocks.

Ghana's economy has suffered due to the over-reliance on these primary traditional exports like the price of gold, cocoa, and oil; which prices have fluctuated internationally in recent years. This highlights the importance of finding new export growth sectors and diversifying exports beyond conventional basic materials (Ghana Statistical Service, 2023). Economic study has focused on the connection between agricultural exports and economic growth, particularly for developing and rising economies. The impact of agricultural exports on economic development has been the subject of numerous studies, with differing findings depending on the region, the kind of agricultural product, and the methodology used.

Murugesan (2019) employed an Error Correction Model on 1990-2017 data and found that both agricultural and non-agricultural exports significantly increase India's real GDP.

To this end, a number of research has determined that a nation's exports directly and favourably correlate with its GDP or economic growth. However, nobody has discussed how NTAEs specifically affect GDP.

In spite of these observations, the literature available does not comprehensively enumerate the government policies that encourage the expansion of NTAEs in order to increase the GDP of developing nations. This is a gap in literature that this study seeks to identify. Thus, the policy measures and best practices that can advance the development of Ghana's non-traditional agricultural exports (NTAEs). This thesis will further outline interventions within the NTAE sector that align with Ghana's commitments under the Global Agenda 2030 and the Sustainable Development Goals (SDGs).

2.5 NTAEs, Value Chains, and Employment in Ghana

Through both direct and indirect value-chain connections, non-traditional agricultural exports (NTAEs) influence employment outcomes. They directly create jobs in export logistics, processing, packaging, aggregation, and primary production. In line with the global value chain dynamics described by Gereffi and Fernandez-Stark (2019), they indirectly increase demand for transport services, certification, cold chain management, and input supply. High-value horticulture and cashew exports have increased wage employment in Ghana, especially for women and young people, according to recent studies (World Bank, 2020; GEPA, 2023). According to McMillan, Rodrik, and Verduzco-Gallo (2014), Africa's structural transformation has frequently been 'growth-reducing', with labour moving away from advanced industries and agriculture and toward low-productivity informal services.

Concerns on whether NTAE expansion is adequately labour-absorbing are raised by Ghana's Labour Force Survey (GSS, 2022), which reveals a steady decline in agricultural employment. These conflicting results highlight the need for context-specific research on the effects of NTAEs on sectoral labour reallocation and employment quality.

According to structural transformation theory (Lewis, 1954; Kuznets, 1966), labour transferring from low-productivity agricultural to higher-productivity sectors is necessary for economic progress. However, this shift has been uneven or sluggish in many African economies (McMillan & Rodrik, 2011; Nin-Pratt, 2021).

Through the creation of labour-intensive horticulture jobs, the stimulation of agro-processing companies, and the expansion of transportation and export services, NTAEs have a unique potential to expedite this process.

According to a 2019 Ghana Statistical Service (GSS) study, the main employer employing approximately 45.38 percent of the workforce is agriculture. The nationwide employment distribution by agriculture industry from 2000 to 2023 is depicted below:

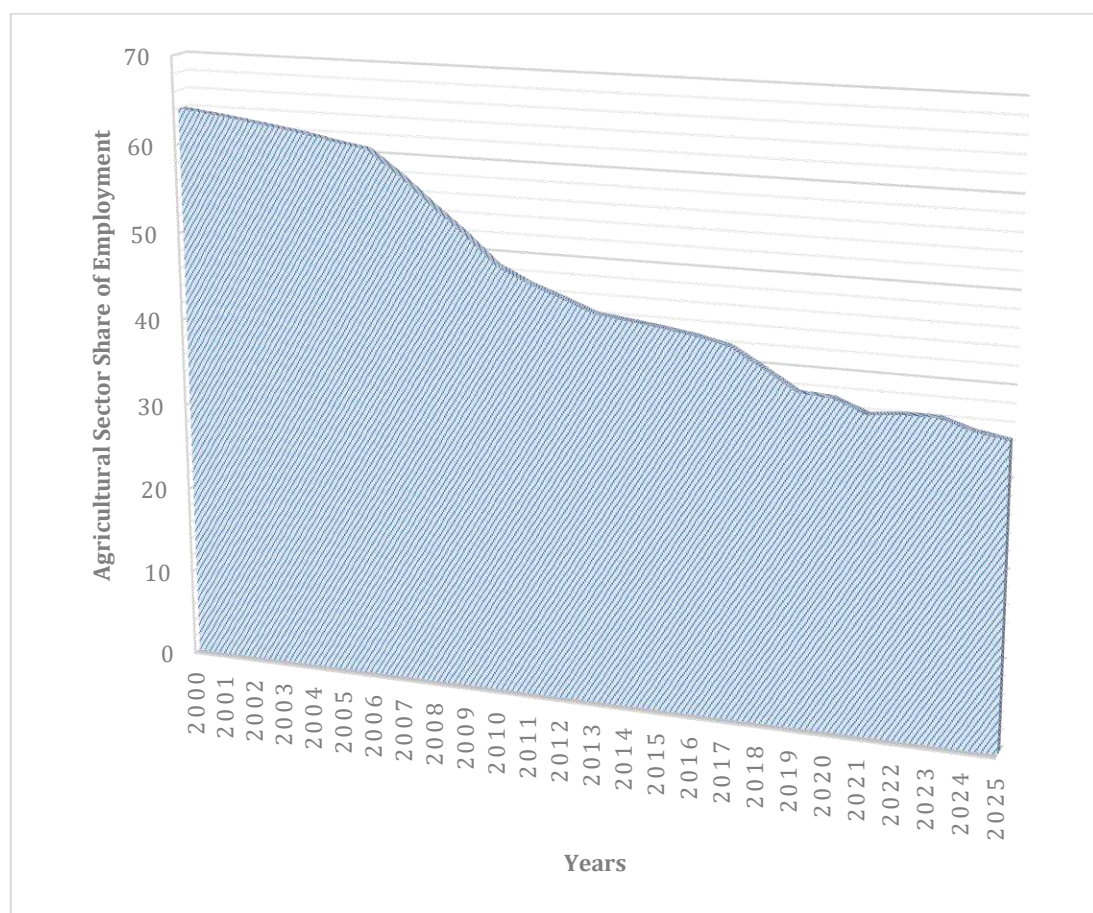


Figure 4 National Distribution Share of Employment by Agricultural Sector (2000-2025)
Source: Author's Design with data from GSS 2023 Labour Force Survey report and World Bank

Over the previous 25 years, Ghana's agricultural workforce structure has significantly changed. The percentage of employment in agriculture fell from 64% in 2000 to 34% in 2025, as the figure above illustrates, indicating larger structural changes in accordance with trends seen throughout Sub-Saharan Africa. Per some current data from the *Journal of African Economies and World Development*, Ghana's labour market is still changing due to mechanization, urbanization, and growing service-sector dynamism (Adjaye-Gbewonyo et al. 2021), Osei & Jedwab, 2023).

The growth of Non-Traditional Agricultural Exports (NTAEs) has produced new types of work throughout expanding value chains, even if the percentage of employment in agriculture is declining. Compared to conventional subsistence farming, these specialized jobs in packing, cold-chain logistics, export certification, processing, and value addition offer higher productivity and more stable wage employment. Research on Ghana's horticulture, cashew, and shea industries confirms that NTAEs are progressively helping young people and women find jobs while promoting auxiliary service industries related to logistics and export compliance (World Bank, 2020; GEPA, 2023; Food Policy, 2022).

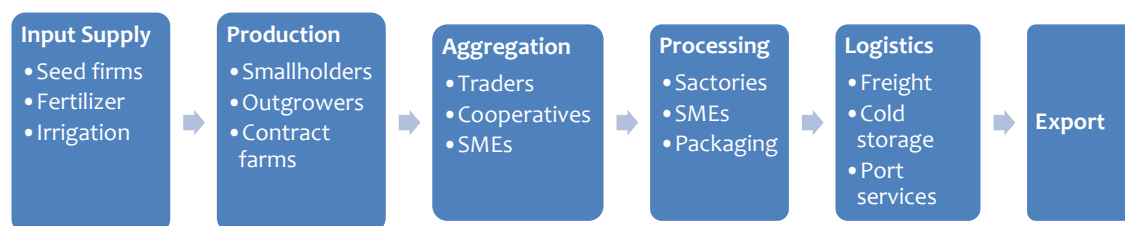


Figure 5 NTAE Value Chain Employment Structure

Source: Author's own Design

Ghana's Non-Traditional Agricultural Exports (NTAEs) play an increasingly important role in employment generation and environmental sustainability debates. Evidence shows that NTAE value chains are major rural livelihood sources, with the cashew sector alone engaging over 300,000 farmers in the Bono and Bono East regions and employing more than 5,000 women in processing facilities, highlighting its gender-inclusive industrial potential (MOFA, 2022). Horticultural exports such as pineapples, mangoes, and vegetables further create substantial seasonal employment for youth in farming and packhouse operations, while also stimulating non-farm rural jobs in logistics, packaging, and certification services. The shea industry is particularly significant, involving over 600,000 women in nut collection and processing across Northern Ghana (GEPA, 2023). Collectively, these figures demonstrate the labour-absorbing capacity of NTAEs across multiple value-chain nodes. However, employment outcomes are not uniformly positive. Seasonal work remains unstable, mechanization reduces labour demand in some crops, and stringent export standards may exclude smallholders. These complexities justify the ambiguous employment hypothesis in existing literature. Empirical assessment is further constrained by limited labour-market data, especially for undocumented and seasonal agricultural workers, creating gaps in quantifying NTAE-related employment effects.

This thesis addresses these gaps using time-series data from 2000-2025 to examine how NTAEs can contribute to reducing Ghana's rising youth unemployment, which reached 14% in 2024 (GSS, 2024). The analysis explores how agricultural trade policies can enhance labour absorption and inclusive growth.

2.6 NTAEs and the Environmental Sustainability of Ghana

Environmental sustainability is equally critical. Ghana faces mounting ecological pressures, including deforestation, land degradation, water stress, and rising CO₂ emissions (World Bank, 2024). Research presents mixed findings: some studies link export-driven agricultural expansion to deforestation and biodiversity loss (Land Use Policy, 2023; World Development, 2022), while others argue that global value-chain participation can promote sustainable production standards (Gereffi & Lee, 2021). Ghana's forest cover declined from 39% in 2000 to 35.2% in 2023, and its 2024 Environmental Performance Index score of 27.7 reflects severe environmental vulnerabilities.

Using the Environmental Kuznets Curve (EKC) as a framework (Grossman & Krueger, 1995; Stern, 2004), this study evaluates whether Ghana's NTAE-driven growth is approaching a sustainability turning point. By integrating ecological-footprint data from 2000-2025, the thesis provides a comprehensive assessment of whether export expansion continues to intensify environmental degradation or is gradually shifting toward more sustainable production practices.

The Marrakesh Declaration (CoP 22, 2016) marked a major step toward sustainable commodity governance in Africa by emphasizing forest conservation while acknowledging agriculture's role in deforestation. Building on this foundation, the Africa Sustainable Commodities Initiative (ASCI) enables African producer countries to establish sustainability guidelines for cocoa, coffee, rubber, and palm oil, balancing livelihood improvement with natural-resource protection. Voluntary sustainability standards, defined by UNCTAD (2023) as private requirements ensuring compliance with economic, social, and environmental criteria, play a central role in this transition.

Blyde et al. argue that exporters targeting high-income markets tend to adopt stronger socio-ecological practices due to consumer preferences for environmentally responsible production (Blyde et al., 2021). While their work focuses on short-term environmental effects, this thesis extends the analysis to long-term outcomes. Ecological footprints are widely recognized as a suitable metric for assessing environmental degradation (Li, 2022). Agricultural exports both processed and unprocessed contribute indirectly to deforestation and soil erosion, prompting scholarly attention to their environmental impacts (Jambor, 2020). Despite Ghana's forestry data identifying agriculture as a major driver of forest loss, few studies isolate the specific role of NTAEs. Moreover, long-term ecological indicators are rarely incorporated. By analysing ecological-footprint data from 2000-2025, this study evaluates whether NTAE-led export diversification aligns with sustainable development trajectories.

The chart below (figure 6) shows Ghana's deforestation rate, agricultural greenhouse gas emissions, and fertilizer use from 2000-2025. Fertilizer use rises sharply after 2010, peaking around 2019-2020 due to the Planting for Food and Jobs initiative, which expanded input subsidies and intensified cultivation (FAO, 2023; IFDC, 2022). Deforestation spikes coincide with agricultural land expansion and cocoa frontier clearing (World Bank, 2021). Greenhouse gas emissions increase gradually, reflecting intensified fertilizer application and livestock growth.

The post-2020 decline aligns with global fertilizer price shocks and reduced subsidies. Overall, the chart illustrates how Ghana's agricultural intensification drives environmental pressure through land conversion and input use (FAOSTAT, 2024).

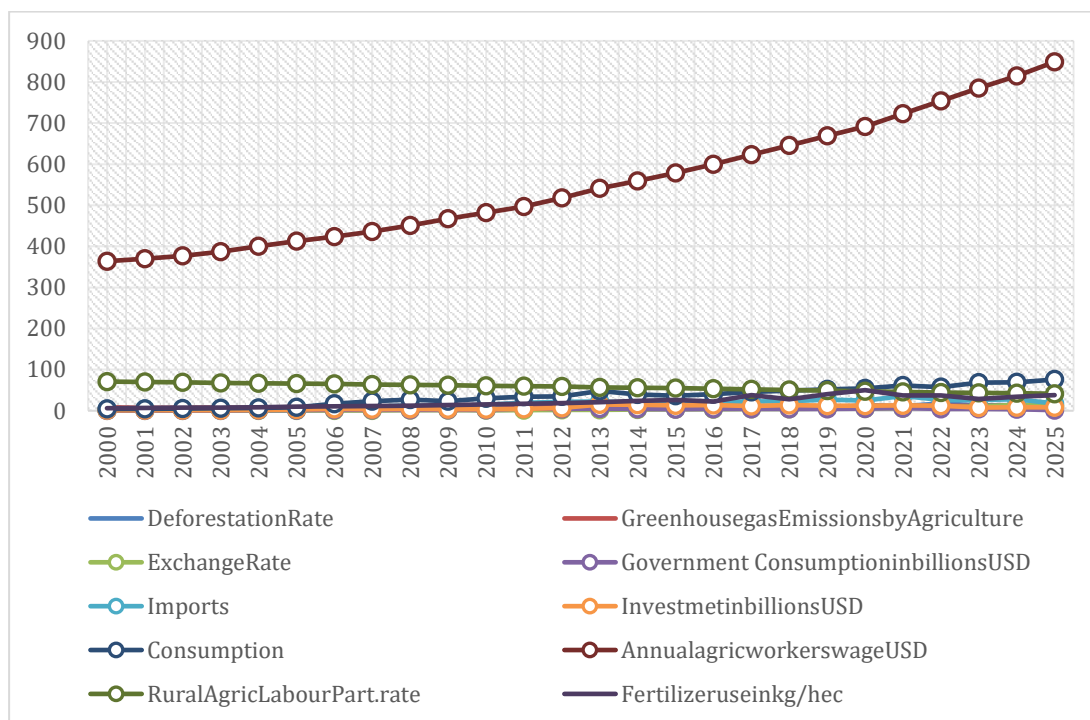


Figure 6 Fertilizer Use, Agricultural Greenhouse Gas Emissions, and Deforestation in Ghana (2000-2025)

Source: Author's Design with data from FAOSTAT, IFDC, World Bank WDI Fertilizer Consumption

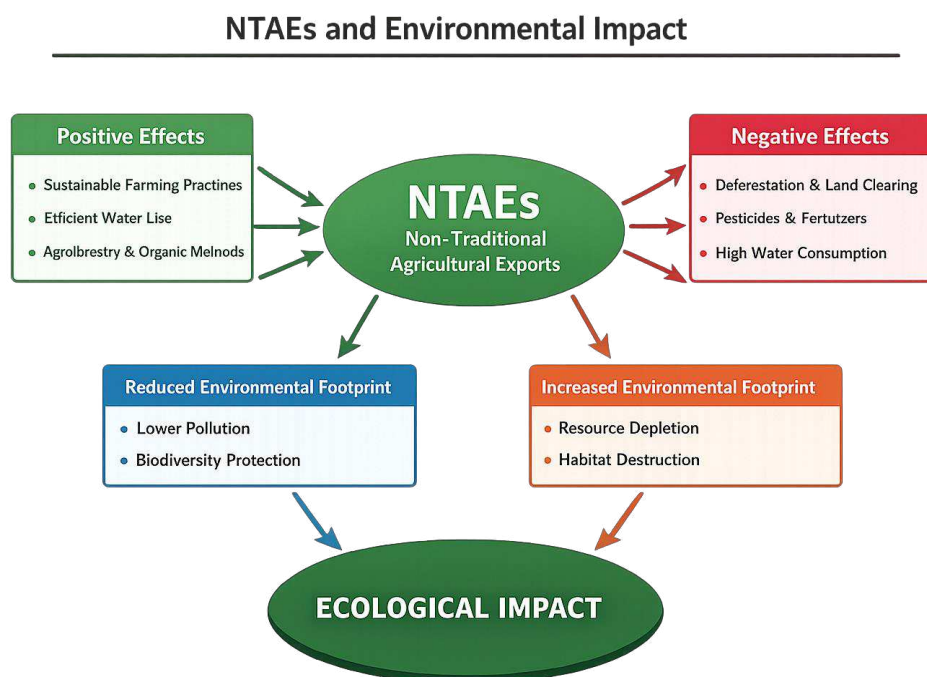


Figure 7 NTAEs and Environmental Impact

Source: Author's Design using copilot prompts

The diagram titled “NTAEs and Environmental Impact” shows how NTAEs affect ecological outcomes through both positive and negative pathways. On the left, sustainable farming, efficient water use, and agroforestry practices reduce pollution and protect biodiversity. On the right, deforestation, chemical use, and high-water consumption lead to resource depletion and habitat destruction. Both pathways ultimately shape Ghana’s overall ecological impact, showing that NTAEs can either enhance environmental sustainability or intensify degradation depending on production practices and resource management.

2.7 Gaps in Existing Literature

The reviewed literature reveals several critical gaps that this thesis addresses. To start with, the existing studies on Ghana’s exports largely employ aggregate data, obfuscating the distinct contributions of Non-Traditional Agricultural Exports (NTAEs) to GDP, employment, and environmental outcomes. Secondly, there is limited longitudinal and econometric evidence isolating NTAE impacts, with existing research relying on descriptive or cross-sectional analyses. More so, the integration of sustainability dimensions particularly ecological footprints and environmental trade-offs remains weak, limiting understanding of how export growth aligns with the SDGs. Fourthly, prior works inadequately explore value-chain linkages and labour dynamics, neglecting gender and youth employment effects. Finally, the published literature is missing a holistic conceptual framework combining Export-Led Growth, Structural Transformation, Global Value Chains, and the Environmental Kuznets Curve. This thesis fills these gaps through a multidimensional, time-series analysis of NTAEs from 2000-2025, linking economic, social, and ecological outcomes empirically.

3 METHODOLOGY

3.1 Introduction

This chapter details the quantitative research design adopted to analyse the socio-economic and ecological impacts of Non-Traditional Agricultural Exports (NTAEs) on Ghana's domestic economy. The methodology aligns with the research questions by focusing on statistical relationships, econometric modelling, and policy simulations using secondary numerical data.

3.2 Research Design

This study employs a quantitative longitudinal research design using time-series data from 2000-2025 to examine how Non-Traditional Agricultural Exports (NTAEs) influence Ghana's GDP, employment, and environmental outcomes. A quantitative approach is appropriate because it enables the measurement of statistical relationships among variables and supports econometric modelling of both short-run fluctuations and long-run trends (Pesaran et al., 2001; Sunde, 2017). The selected period captures recent developments in Ghana's export diversification strategy and allows for dynamic analysis rather than static comparisons, consistent with prior studies using similar designs in developing economies (Awokuse, 2008; Hye & Lau, 2015; Shahbaz et al., 2014). This period reflects recent developments in Ghana's export diversification strategy and provides a relevant basis for policy analysis. This approach is suitable for examining dynamic relationships, rather than static associations. It also supports the use of econometric techniques such as the Autoregressive Distributed Lag (ARDL) model, which requires time-series data to estimate short- and long-term effects. The choice of a quantitative design aligns with the objectives of the study, which are to: measure the relationship between NTAEs and macroeconomic indicators such as GDP, and employment; apply econometric models to analyse both short-term and long-term effects; provide empirical evidence to support policy decisions on export diversification and sustainability. In addition, the use of secondary data from reliable sources improves the objectivity and consistency of the analysis. It also reduces bias and allows the results to be verified. The application of statistical methods further strengthens the study by moving beyond description to evidence-based conclusions. The research design is also consistent with the export-led growth theory, which suggests that increased exports contribute to economic growth. Using time-series data, the study can test this relationship within Ghana's NTAE sector. To ensure the study reflects the most recent economic and environmental dynamics in Ghana, the dataset was extended to include 2024 and 2025 observations where available. While 2025 data remain partially provisional, they are incorporated to provide updated insights and enhance the policy relevance of the study. The inclusion of these recent periods strengthens the robustness of the time-series analysis and enables assessment of post-2023 macroeconomic stabilization efforts, particularly regarding employment trends, export performance, and environmental pressures.

3.3 Data Sources

The study employs annual time-series data covering 2000-2025, sourced from credible secondary institutions such as the Ghana Statistical Service (GSS), the World Bank, and related national and international databases. Although the dataset provides valuable insights into trends in NTAEs and key macroeconomic indicators, the relatively small sample size necessitates a parsimonious modelling strategy and limits the use of highly data-intensive econometric techniques. As a result, the findings emphasize statistical associations rather than definitive causal claims.

Primary data sources include GSS records on economic indicators, agricultural exports, and employment. The analysis also incorporates updated datasets for 2024 and 2025 from GEPA, GSS, and the Bank of Ghana, which enhance the relevance of the study by reflecting recent developments in export performance, employment patterns, GDP growth, and environmental conditions. The inclusion of these recent observations strengthens the robustness of the time-series analysis and improves the policy relevance of the results.

3.4 Data Collection Procedure

The study employed a structured secondary data collection process to ensure accuracy and consistency. Annual time-series data covering 2000-2025 were systematically identified, retrieved, and organized to align with the research objectives. Key variables economic indicators, employment measures, and environmental metrics were selected for their relevance to assessing the impact of Non-Traditional Agricultural Exports (NTAEs).

Data were obtained from credible institutions, including the World Bank, Ghana Export Promotion Authority (GEPA), Ghana Statistical Service (GSS), and Environmental Protection Agency (EPA) Ghana. All datasets were downloaded from official sources, verified for reliability, and cleaned in Microsoft Excel to address missing values, correct inconsistencies, standardize units, and ensure comparability across years, enhancing validity for econometric analysis.

Table 3 Time-Series Data of NTAEs, Economic indicators (GDP, Imports, Employment), and Environment Indicators (2000-2025)

YEAR	Real NTAE (USD Billion)	Other NTE (USD Billion)	Real Trad. Exports (USD Billion)	Real GDP (USD Billion)	Imports (USD Billion)	Agric. Empt. (% of Total Emp.)	Def. Rate (Primary Forest lost in kha)	GH gas Emissions by Agric (Mt)
2000	0.08	0.70	25.13	53.09	35.68	64.00	0.70	8.80
2001	0.08	0.74	24.14	55.22	35.78	63.57	0.50	7.13
2002	0.09	0.78	23.71	57.7	31.68	63.09	4.4	6.98
2003	0.1	0.84	23.76	60.7	34.36	62.57	1.9	7.73
2004	0.1	0.92	24.17	64.1	38.72	62.01	1.6	7.22
2005	0.11	1.00	23.60	67.88	41.88	61.31	3.4	8.94
2006	0.12	0.77	9.13	72.23	16.19	60.72	3.6	8.60
2007	0.12	1.04	10.68	75.37	19.70	58.08	3.0	8.15
2008	0.19	1.15	12.48	82.26	24.56	55.00	2.7	8.21
2009	0.15	1.07	13.66	86.25	21.47	52.11	3.5	7.95

2010	0.17	1.46	16.91	93.06	27.21	48.84	1.6	8.39
2011	0.3	2.12	26.10	106.13	38.11	47.07	2.8	9.25
2012	0.28	2.08	30.49	116	42.99	45.76	2.5	8.78
2013	0.32	2.12	29.23	124.48	43.96	44.34	5.1	9.59
2014	0.34	2.17	28.86	128.03	37.63	43.89	7.4	9.42
2015	0.4	2.12	34.37	130.75	40.61	43.42	3.6	9.88
2016	0.37	2.09	35.50	135.16	40.17	42.86	14	9.91
2017	0.44	2.12	44.43	146.15	43.25	42.01	7.7	10.4
2018	0.59	2.22	47.29	155.21	45.40	39.89	12	10.0
2019	0.43	2.47	53.58	165.31	52.60	37.65	5.8	10.3
2020	0.43	2.42	46.41	166.16	47.98	37.41	14	11.8
2021	0.48	2.85	45.34	174.59	50.76	35.96	11	10.8
2022	0.51	3.02	45.24	181.24	51.47	36.32	18	12.1
2023	0.49	3.45	47.33	186.93	50.86	36.32	13	11.3
2024	0.52	3.31	52.09	197.38	55.71	35.18	14	11.6
2025*	0.71	3.34	77.42	207.83	74.82	34.57*	14.5*	11.9*

Source: Author's Design with data from GEPA 2015-2024 Annual report, Statista, GSS, World Bank, MOFA, FAO, EPA, Energy Commission, Ghana, and Global Forest watch, etc.

Some of the 2025* data are interpolations from sources used as official verified data has not been published. Also, NTAE, Other NTE, Traditional Exports, Imports, and GDP values were at **constant 2015 prices**.

3.5 Data Extraction Instrument

Data extraction was carried out using Microsoft Excel, which served as the primary tool for compiling and organizing the dataset. A structured spreadsheet template was designed to ensure consistency in recording and managing the data.

The dataset was arranged in columns representing each variable, including NTAE revenue, GDP, inflation, employment, and environmental indicators. Rows were used to capture annual observations for the period 2000-2025. This structure allowed for easy tracking of trends and ensured compatibility with econometric software used for analysis. To enhance accuracy, the extracted data were carefully reviewed and cross-checked against sources. The dataset was also standardized to maintain uniform measurement units across all variables.

3.6 Data Analysis

Data analysis was conducted using Gretl, and EViews 12, both widely recognized econometric software packages suitable for advanced time-series analysis. The study employed a combination of econometric techniques to examine the relationships between Non-Traditional Agricultural Exports (NTAEs) and selected economic and environmental indicators. The analysis was carried out in three stages. First, descriptive statistics were used to summarize the data and identify general trends over the study period. This provided an initial understanding of the behaviour of key variables such as NTAE revenue, GDP, inflation, employment, and environmental indicators.

Second, the Autoregressive Distributed Lag (ARDL) model was applied to estimate both short-run and long-run relationships between variables. The ARDL approach was selected because it accommodates variables with mixed orders of integration and performs well with small sample sizes. It also allows for the simultaneous estimation of dynamic adjustments and long-term equilibrium relationships.

Third, Error Correction Model (ECM) models were used to assess the short-term impact of NTAEs on GDP and employment, while ensuring adjustment toward a long-run equilibrium relationship among. This approach helps isolate the specific contribution of NTAEs to economic performance.

3.7 Model Formulation

The specification of the econometric models in this study is grounded in established economic theories rather than purely data-driven assumptions. Specifically, the models are anchored in the Export-Led Growth (ELG) theory, Structural Transformation theory, and the Environmental Kuznets Curve (EKC).

The Export-Led Growth hypothesis posits that export expansion contributes to economic growth through increased foreign exchange earnings, productivity spillovers, and economies of scale. In this context, Non-Traditional Agricultural Exports (NTAEs) are expected to have a positive effect on Ghana's GDP.

The Structural Transformation theory explains how labour shifts across sectors as economies develop. The expansion of NTAEs is expected to influence employment through value-chain activities, including farming, processing, packaging, and export logistics.

The Environmental Kuznets Curve (EKC) provides the theoretical basis for analysing the relationship between export expansion and environmental outcomes. It suggests that economic activities may initially increase environmental degradation but may later improve environmental quality as economies adopt sustainable practices. Therefore, the models specified in this study are theory-driven and reflect the economic, employment, and environmental transmission channels through which NTAEs affect the domestic economy.

3.7.1 Model 1: NTAEs and GDP

Theoretical Rationale

This model is to assess the impact of non-traditional agricultural exports (NTAEs) on Ghana's economic growth. It is to estimate how Ghana's GDP responds to NTAEs while controlling for key macroeconomic variables. While NTAEs capture the export-led growth effect, other variables like imports represent leakages from the circular flow, government expenditure reflects fiscal stimulus, investment captures capital accumulation, and consumption represents domestic demand.

The exchange rate accounts for external competitiveness and the cost of trade. By including these variables simultaneously, the model isolates the independent contribution of NTAEs to GDP growth while all other regressors are held constant.

The method combines time-series econometric approaches appropriate for emerging economies, modern macroeconomic theory, and national income accounting principles.

The goal is to create a robust empirical framework that can capture the long-term equilibrium effects of NTAEs on GDP as well as their short-term dynamics.

GDP represents the total value of goods and services produced within an economy. In open-economy macroeconomics, using the ELG theory, the GDP function can be expressed as:

$$GDP_t = \beta_0 + \beta_1 NTAE_t + \beta_2 TE_t + \beta_3 M_t + \beta_4 G_t + \beta_5 I_t + \beta_6 C_t + \varepsilon_t \quad (1)$$

Where;

$NTAE_t$ = Non-Traditional Agricultural exports at period t

TE_t = Traditional Exports

C_t = Household Consumption

I_t = Investment or Capital formulation

G_t = Government Expenditure

M_t = Imports of capital goods, machinery, inputs and technology

ε_t = Error term

The model specifies GDP as a function of non-traditional exports, traditional exports, and productive imports, Government expenditure, Capital formation or Investments, and Household Consumption; consistent with Export-Led Growth theory. Coefficients measure each variable's contribution to output. Positive estimates imply growth-enhancing effects, while the error term captures unobserved influences. The model evaluates export diversification and import-driven productivity effects. Contemporary economic theory argues that export expansion stimulates GDP through productivity spillovers, foreign exchange earnings, economies of scale, and structural transformation (Krugman, Obstfeld, & Melitz, 2018; Hausmann, Hwang, & Rodrik, 2007). Non-Traditional Agricultural Exports are part of the total Exports that is accounted for in the computation of GDP and it has a positive influence on GDP as depicted in the econometric model above. The Export-Led Growth (ELG) hypothesis further posits that economies with strong export sectors tend to experience faster and more sustained growth. Empirical evidence from Asia and Sub-Saharan Africa shows that export diversification and value addition significantly enhance GDP performance (Awokuse, 2008; Hye & Lau, 2015; Sunde, 2017). For Ghana, where traditional exports such as cocoa, gold, timber, and crude oil dominate, the rise of NTAEs presents an opportunity to broaden the export base and reduce vulnerability to commodity price shocks.

Econometric Methods used for Analysing GDP and NTAEs

GDP and exports are examples of time-series macroeconomic variables that frequently show non-stationarity, structural breakdowns, and long-run co-movement.

Pesaran, Shin, and Smith's (2001) Autoregressive Distributed Lag (ARDL) modelling approach is used in this study to address these features. The reasons the ARDL technique is so appropriate include;

- It supports variables that are integrated of order I (0) and I (1).
- Small sample sizes are typical in datasets from underdeveloped nations, and it works well with them.
- It makes a clear distinction between effects that happen in the short and long term.
- By using lag regressors, endogeneity issues are lessened.

The Augmented Dickey-Fuller (ADF), Phillips-Perron (PP), and KPSS procedures are used to perform unit root tests to ascertain the integration order of each variable prior to estimating the ARDL model.

Basic ARDL specification for the link between NTAEs and GDP

The general ARDL (p, q₁, q₂, q₃, q₄, q₅, q₆) form and Error correction is shown below:

$$\text{GDP}_t = \alpha_0 + \sum_{i=1}^p \alpha_i \text{GDP}_{t-i} + \sum_{j=0}^{q_1} \beta_j \text{NTAE}_{t-j} + \sum_{k=0}^{q_2} \gamma_k \text{TE}_{t-k} + \sum_{l=0}^{q_3} \delta_l \text{M}_{t-l} + \sum_{m=0}^{q_4} \theta_m \text{G}_{t-m} + \sum_{n=0}^{q_5} \phi_n \text{I}_{t-n} + \sum_{r=0}^{q_6} \psi_r \text{C}_{t-r} + \varepsilon_t \quad (2)$$

Where:

GDP_t = Gross Domestic Product at time t (Dependent Variable)

α_0 = Constant term - the baseline GDP when explanatory variables are zero

$\sum_{i=1}^p \alpha_i \text{GDP}_{t-i}$ = Current and Lagged GDP terms - Captures persistence or inertia in economic growth (past GDP influences current GDP)

$\sum_{j=0}^{q_1} \beta_j \text{NTAE}_{t-j}$ = Current and Lagged NTAE terms-Assess the short and long-run effects of NTAEs on GDP

$\sum_{k=0}^{q_2} \gamma_k \text{TE}_{t-k}$ = Current and Lagged Traditional Exports - Captures the contribution of traditional exports (e.g., cocoa, gold, timber, and crude oil) to GDP.

$\sum_{l=0}^{q_3} \delta_l \text{M}_{t-l}$ = Current and Lagged Imports - Depicts the influence of import levels on domestic output often negative due to leakage effects.

$\sum_{m=0}^{q_4} \theta_m \text{G}_{t-m}$ = Current and lagged Government Expenditure - This captures the role of public spending as a driver of economic activity.

$\sum_{n=0}^{q_5} \phi_n \text{I}_{t-n}$ = Current and lagged Investment - This is to measure the effect of capital formation on the economic growth of Ghana.

$\sum_{r=0}^{q_6} \psi_r \text{C}_{t-r}$ = Current and lagged household Consumption - This term reflects the contributions of household demand to GDP

ε_t = Error term - Captures random shocks and unobserved influences on GDP.

The ARDL model explains how Ghana's GDP responds dynamically to its own past values and lagged effects of exports and imports. It captures both short-run adjustments and long-run equilibrium relationships. NTAEs, traditional exports, investment, household, public expenses drive growth, while imports may dampen it, reflecting export-led development and macroeconomic interdependence.

Long-run (cointegrating) relationship:

The ARDL in its Error Correction Model (ECM) form shows;

$$\Delta GDP_t = \lambda ECT_{t-1} + \sum_{i=1}^{p-1} \phi_i \Delta GDP_{t-i} + \sum_{j=0}^{q1} \beta_j \Delta NTAE_{t-j} + \sum_{k=0}^{q2} \gamma_k \Delta TE_{t-k} + \sum_{l=0}^{q3} \delta_l \Delta M_{t-l} + \sum_{m=0}^{q4} \theta_m \Delta G_{t-m} + \sum_{n=0}^{q5} \phi_n \Delta I_{t-n} + \sum_{r=0}^{q6} \psi_r \Delta C_{t-r} + \varepsilon_t \quad (3)$$

Where:

ΔGDP_t = Change in GDP - It shows short-run fluctuations in economic growth.

$\lambda ECT_{t-1} = \lambda(GDP_{t-1} - \theta_1 NTAE_{t-1} - \theta_2 TE_{t-1} - \theta_3 M_{t-1} - \theta_4 G_{t-1} - \theta_5 I_{t-1} - \theta_6 C_{t-1})$
Error-correction term - It long-run relationship among GDP and its determinants.

$\sum_{i=1}^{p-1} \phi_i \Delta GDP_{t-i}$ = Short-run GDP dynamics - It reflects the effect if past changes in GDP on current GDP growth.

$\sum_{j=0}^{q1-1} \phi_j \Delta NTAE_{t-j}$ = Short-run impact of NTAE - It represents the immediate influence of NTAE on economic performance.

$\sum_{k=0}^{q2-1} \omega_k \Delta TE_{t-k}$ = It captures the short-run effects of changes in Traditional Export (TE) on GDP growth.

$\sum_{m=0}^{q3-1} \xi_m \Delta M_{t-m}$ = The effect of Imports (M) - Measures the short-run impact of changes in Imports on GDP growth.

ε_t = Error term - It indicates the random shocks and unobserved influences on GDP

The error-correction model captures the dynamic adjustment of GDP growth by integrating short-run influences of lagged GDP, NTAE, TE, G, C, I and M with a long-run equilibrium framework, wherein the error-correction term quantifies the speed at which deviations from long-run economic equilibrium are progressively corrected over time. This supports the export-led growth hypothesis, showing that NTAE expansion drives sustained economic performance. Overall, the result highlights the need for export competitiveness and diversification to maintain macroeconomic stability and long-term growth.

3.7.2 Model 2: NTAEs and Employment

This model is grounded in labour-demand theory and the export-led employment hypothesis. Employment in the model is explained through labour-demand theory, which posits that firms hire workers based on output expansion and the cost of labour. As NTAEs increase, production and value-chain activities expand, raising the demand for labour in line with the export-led employment hypothesis.

Wages capture labour costs, where higher wage levels may reduce firms' hiring capacity, while interest rates reflect capital costs that influence investment and job creation. Together, the model evaluates how export growth, labour costs, and financial conditions jointly shape employment outcomes in Ghana's NTAE sector. Employment under this study is not about the number of workers but a labour-absorption outcome of export diversification.

NTAEs affect employment through three (3) main channels namely;

- Direct farm jobs: This involves the labour used in NTAE crop production (cashew, shea nuts, palm nut, rubber, yam, vegetables, horticulture, etc.
- Value-chain jobs: Employment that comes as a result of aggregation, processing, packaging, certification, transport, and logistics.
- Spillover jobs: This is the indirect employment generated through higher rural incomes, where earnings from NTAE production and processing stimulate additional jobs via increased local spending and reinvestment.

Theoretical Rationale

The model examines the relationship between Non-Traditional Agricultural Exports (NTAEs) and employment in Ghana's agricultural sector. Recognizing that NTAEs are labour-intensive and often involve smallholder farmers and rural communities, this model aims to quantify the employment effects of expanding NTAE production and export activities. The employment impact of NTAEs is grounded in structural transformation theory and labour economics.

Expanding NTAE production can generate direct employment through farm labour, processing, and logistics, as well as indirect employment via supply chains and service sectors (Barrett et al., 2012; Minten et al., 2018). Given the labour-intensive nature of many NTAEs such as cashew, shea, and pineapples, growth in this sector is expected to have significant positive effects on rural employment and poverty reduction.

The employment function is thereby given as;

$$EMP_t = \beta_0 + \beta_1 NTAE_t + \beta_2 W_t + \beta_3 R_t + \varepsilon_t \quad (4)$$

Wherein;

EMP_t = Employment to population ratio

$NTAE_t$ = Value/Volume of Non-Traditional Agricultural Exports

W_t = Average wage rate in agriculture (USD or local currency)

R_t = Rural labour force participation rate (%)

ε_t = Error term

Employment is modelled as a function of NTAEs, overall economic activity, labour costs, and other structural factors. The underlying hypothesis is that expansion in

NTAEs increases labour demand both on farms and along export value chains, thereby raising employment in the long run.

The model evaluates the subsequent hypotheses:

- The expansion of NTAEs positively impacts agricultural employment.
- Elevated average earnings are associated with enhanced employment incentives.
- Increased participation of the rural labour force enhances employment rates.

ARDL Employment Model (Levels Form)

$$EMP_t = \beta_0 + \beta_1 EMP_{t-1} + \beta_2 NTAE_t - \beta_3 W_t + \beta_4 INF_{t-1} - \beta_5 R_t - \beta_6 GDP_t + \beta_7 GDP_{t-1} + \varepsilon_t \quad (5)$$

Wherein;

EMP_t is the Employment-to-population ratio at time t (dependent variable). It measures labour absorption in the economy.

β_0 : Constant term, representing baseline employment conditions.

EMP_{t-1} : Lagged employment ratio. The positive coefficient ($\beta_1 > 0$) captures employment persistence, reflecting labour-market rigidity and slow adjustment over time.

$NTAE_t$: Non-Traditional Agricultural Exports. A positive coefficient ($\beta_2 > 0$) indicates that export diversification increases labour demand through farming, agro-processing, logistics, and trade-related services.

W_t : Agricultural workers' wage rate. The negative sign ($-\beta_3$) reflects labour-demand theory: higher wages raise production costs and may reduce firms' ability to hire.

INF_{t-1} : Lagged inflation. A positive coefficient ($\beta_4 > 0$) suggests delayed demand-side effects, where past inflation stimulates employment through short-run output expansion.

R_t : Rural agricultural labour participation rate. The negative sign ($-\beta_5$) indicates disguised unemployment or surplus labour in rural agriculture, which lowers employment intensity.

GDP_t : Current GDP. The negative sign ($-\beta_6$) reflects jobless growth or productivity-driven expansion that does not immediately create jobs.

GDP_{t-1} : Lagged GDP. A positive coefficient ($\beta_7 > 0$) shows that economic growth translates into employment with a time lag.

ε_t : Error term, capturing unobserved shocks and omitted factors.

The ARDL model integrates export-led employment, structural transformation, and dynamic adjustment. It shows that NTAEs raise employment directly, while growth effects materialize with delays, and rural labour surplus and rising wages constrain job creation.

3.7.3 Model 3: NTAEs and Environment

The environmental model is grounded in the Environmental Kuznets Curve (EKC), which explains the relationship between economic activity and environmental degradation. The EKC suggests that export expansion, with focus on NTAEs, may initially increase environmental pressure due to increased production (scale effect), but over time may promote sustainable practices (technique effect).

Autoregressive Distributed Lag (ARDL) Model

The ARDL approach is used to capture both short-run and long-run dynamics between NTAEs and environmental indicators. This approach is widely used in time-series analysis due to its flexibility and suitability for small sample sizes.

The ARDL model is appropriate for this study because it accommodates variables with mixed orders of integration, meaning they can be stationary at level I (0) or at first difference I (1).

This eliminates the strict requirement for all variables to be integrated at the same order, which is a limitation of other cointegration techniques.

In addition, the ARDL framework allows for the simultaneous estimation of short-term dynamics and long-term equilibrium relationships. This makes it particularly useful for analysing how changes in NTAEs affect environmental outcomes over time, both in the short run and in the long run.

ARDL (p, q) specification for this model is;

$$\text{EcoFootprint}_t = \alpha_0 + \sum_{i=1}^p \phi_i \text{EcoFootprint}_{t-i} + \sum_{j=0}^q \theta_j \text{NTAE}_{t-j} + \varepsilon_t \quad (6)$$

Where;

EcoFootprint_t = environmental indicator such as ecological footprint, CO₂ emissions from agriculture, deforestation index, agricultural land pressure, etc.

NTAE_t = Non-traditional agricultural exports at time t-j

ϕ_i = -lag coefficients on lagged environmental variables

θ_j = lag coefficients on current and lagged NTAEs

ε_t = error term-it covers shocks, omitted factors, measurement error, etc.

The model incorporates lagged values of both the dependent and independent variables to capture dynamic adjustments. The presence of a long-run relationship was tested using the bounds testing approach, where the calculated F-statistic is compared with critical values. If the statistic exceeds the upper bound, a long-run relationship is confirmed.

The ARDL in its Error Correction Model (ECM) form shows;

$$\Delta \text{EcoFootprint}_t = \alpha + \sum_{j=1}^p \gamma_j \Delta \text{EcoFootprint}_{t-j} + \sum_{j=0}^q \delta_j \Delta \text{NTAE}_{t-j} + \lambda \text{EcoFootprint}_{t-1} + \beta \text{NTAE}_{t-1} + \varepsilon_t \quad (7)$$

Where:

$\Delta \text{EcoFootprint}_t$ = First difference of ecological footprint at time t (dependent variable).

α = Constant term - It captures the baseline or average change when all other variables are zero

$\sum_{j=1}^p \gamma_j \Delta \text{EcoFootprint}_{t-j}$ = lagged differences of ecological footprint ($\Delta \text{EcoFootprint}_{t-j}$).

$\sum_{j=0}^q \delta_j \Delta \text{NTAE}_{t-j}$ = lagged changes in NTAEs (ΔNTAE_{t-j}).

$\lambda \text{EcoFootprint}_{t-1}$ = Lagged level of ecological footprint in the long-run.

βNTAE_{t-1} = Lagged level of NTAE. It measures long-run influence of NTAEs on ecological footprint.

ε_t = White-noise error term, assumed to be independently and identically distributed.

The model combines short-term adjustments (γ_i and δ_j) with long-term equilibrium (via λ and β). To check the diagnostic and robustness of the results, Augmented Dickey-Fuller model will be used to test for stationarity. The bounds test (Pesaran et al., 2001) was used to evaluate cointegration by testing the null hypothesis $H_0: \lambda = \beta = 0$. If the computed F -statistic exceeds the critical upper bound, a stable long-term relationship exists. The model provides empirical evidence for balancing export diversification with environmental sustainability, designing green export policies under Ghana's NTE strategy, Integrating environmental indicators into trade performance metrics, etc.

Table 4 Hypothesis Analysis

Hypothesis	Narration	Expected Sign
H₁	There exists a long-run equilibrium relationship between ecological footprint and NTAEs.	$\lambda < 0$ (significant)
H₂	Changes in NTAEs have a short-run effect on ecological footprint.	$\delta_j \neq 0$
H₃	NTAE expansion increases ecological footprint in the long run.	$\beta > 0$
H₄	The system adjusts toward equilibrium after short-run shocks.	$\lambda < 0$

Source; Author's design using copilot

Empirical Implications

If the error-correction coefficient $\lambda = -0.42$ ($p < 0.01$), this indicates that approximately 42% of the previous year's disequilibrium is corrected annually, confirming a stable long-run relationship and validating the ECM structure. A positive long-run coefficient, for example $\beta = 0.27$ ($p < 0.05$), implies that a 1% increase in NTAEs is associated with a 0.27% rise in environmental pressure, suggesting that export expansion intensifies ecological stress through scale effects.

Conversely, if the estimated coefficient is negative, such as $\beta = -0.18$ ($p < 0.10$), it would indicate that NTAEs reduce environmental degradation, possibly due to adoption of eco-friendly production methods, compliance with sustainability standards, or increased value-added processing. These numerical outcomes guide policy interpretation by distinguishing whether NTAE-led growth is environmentally enhancing or environmentally burdensome.

3.8 Variable Definition and Measurement

To ensure consistency and empirical validity, all variables are clearly defined and aligned with available data.

Non-Traditional Agricultural Exports (NTAEs)

NTAEs represent the value of agricultural export products that are not traditionally dominant export commodities. These typically include horticultural products, processed agricultural goods, and other emerging export crops. The variable is measured as the total value of NTAEs (in monetary terms), serving as the key independent variable.

Economic Growth (GDP)

Economic growth, commonly proxied by Gross Domestic Product (GDP), reflects the sustained expansion of an economy's output of goods and services over time (World Bank, 2023).

Gross Domestic Product (GDP) is the total market value of all final goods and services produced within a country's borders over a specified period, and it remains the most widely used indicator of national economic performance (IMF, 2024). It provides a standardized metric for comparing production over time and between nations by aggregating economic activity through the expenditure components of consumption, investment, government spending, and net exports (World Bank, 2023). Because it represents the scope of economic activity and directs monetary and fiscal policy choices, GDP is essential to macroeconomic analysis (OECD, 2023). It is still the main standard for evaluating economic growth and development despite its drawbacks, such as the exclusion of non-market activity and income distribution (WEF, 2024). GDP reflects the overall economic performance and serves as a primary dependent variable in the analysis.

Employment (EMP)

Employment under this study is the engagement of individuals in productive economic activities that generate income, whether through wage labour, self-employment, or informal work arrangements (ILO, 2024). In developing economies, agriculture remains a major source of employment, absorbing a large share of the labour force despite ongoing structural transformation. Recent evidence shows that agriculture continues to employ over one-third of workers in Sub-Saharan Africa, underscoring its role in rural livelihoods and poverty reduction (World Bank, 2023). The FAO (2024) further notes that agricultural employment is increasingly shaped by commercialization and value-chain expansion, particularly in high-value export crops.

In Ghana, agriculture still accounts for a significant share of national employment, even as labour gradually shifts toward services and industry (GSS, 2023). Thus, employment in agriculture remains central to economic development, labour absorption, and export-driven growth.

Environmental Indicator (ENV)

Environmental impact is proxied using carbon dioxide (CO₂) emissions, measured in metric tonnes. This variable captures the environmental implications of economic and export activities.

3. 9 Integration of Models

Collectively, these models address the study's multifaceted objectives:

The ARDL model evaluates the sustainability of NTAEs by linking export activities to ecological footprints, critical for achieving SDG 12 (Responsible Consumption and Production).

It was also used to quantify the immediate economic benefits, informing policies to enhance NTAE contributions to GDP (SDG 8: Decent Work and Economic Growth). The ARDL was used for measuring the linkages between NTAEs and agricultural employment. Overall, the empirical models are derived from the study's conceptual framework. Each model represents a specific channel, economic, labour, and the environment through which Non-Traditional Agricultural Exports (NTAEs) affect GDP, employment, and ecological outcomes. This ensures the analysis is grounded in theory rather than based on arbitrary relationships.

3.10 Data Management and Validation

The datasets were pre-processed to ensure consistency and analytical validity. NTAE revenues were converted from GHS to USD using Bank of Ghana exchange rates, and GDP and NTAE values were deflated with CPI to obtain constant 2015 USD. Missing employment data were linearly interpolated, and logarithmic transformations were applied to non-stationary variables. Stationarity was assessed using ADF tests, followed by ARDL bounds testing to establish cointegration. Model robustness was evaluated through alternative lag structures and sensitivity checks. These procedures ensured reliable estimation of both short-run dynamics and long-run relationships for sound policy interpretation.

3.11 Unit Root and Cointegration Testing

Before estimation, unit root tests are conducted to determine the order of integration of the variables. This ensures that none of the variables are integrated of order I (2), which would invalidate the use of the ARDL approach.

The existence of a long-run relationship among the variables is examined using the ARDL bounds testing procedure. This approach determines whether the variables are cointegrated, thereby justifying the estimation of long-run coefficients.

3.12 Diagnostic Tests

To ensure the reliability and robustness of the estimated model, diagnostic tests are conducted. These include tests for serial correlation and model stability.

The results of these tests confirm that the model satisfies the necessary econometric assumptions and is suitable for inference within the study's constraints.

3.13 Methodological Framework

The study adopts a conceptual framework linking Non-Traditional Agricultural Exports (NTAEs) to Ghana's economic performance. The framework identifies NTAEs as the independent variable, economic growth (GDP) and employment as dependent variables, and macroeconomic factors such as inflation and exchange rates as control variables. This framework provides a visual and theoretical guide for understanding the relationships examined in this study.

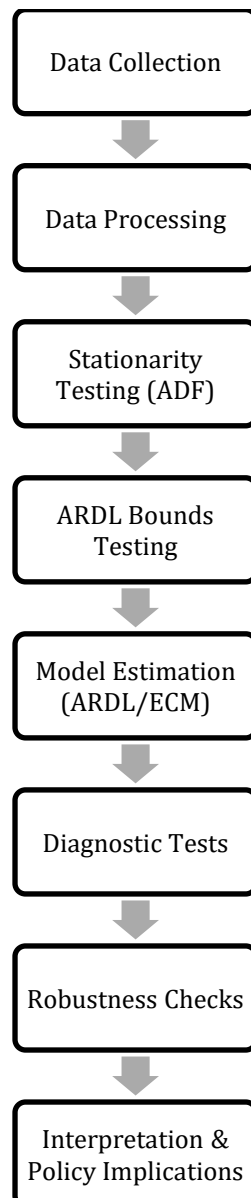


Figure 8 Methodological Framework diagram
Source: Author's design with copilot prompts

3.14 Research Hypotheses

NTAE Export Growth and Economic Performance

H₀: There is no significant relationship between NTAE export growth and Ghana's economic performance.

H₁: There is a significant relationship between NTAE export growth and Ghana's economic performance.

H₀: NTAE exports do not cause Granger-cause changes in GDP.

H₁: NTAE exports Granger-cause changes in GDP

NTAEs and Employment

H₀: There is no significant relationship between NTAE export growth and employment in Ghana.

H₁: There is a significant relationship between NTAE export growth and employment in Ghana.

NTAEs and Socio-Ecological Environment

H₀: There is no significant relationship between NTAE export growth and the socio-ecological environment in Ghana.

H₁: There is a significant relationship between NTAE export growth and the socio-ecological environment in Ghana.

3.15 Operationalization of Variables

The variables in this study are operationalized to ensure clarity and consistency in measurement. Non-Traditional Agricultural Exports (NTAEs) are measured in terms of both metric tons and export value in United States dollars (USD), capturing the quantity and economic significance of exports. Economic growth (GDP) is measured as the annual percentage change in real GDP, reflecting the overall performance of the national economy. Employment is captured by the number of jobs generated within the agricultural export sector, serving as a proxy for labour absorption linked to NTAE activities. In addition, the study incorporates control variables, including the inflation rate (expressed as a percentage) and the exchange rate (USD), to account for macroeconomic factors that may influence the relationships under investigation.

3.16 Methodological Considerations and Limitations

While the study employs appropriate econometric techniques, several limitations are acknowledged. First, the relatively small sample size limits the generalizability of the findings and constrains the use of more advanced econometric models.

Second, a potential issue of endogeneity may arise due to the possibility of bidirectional relationships between NTAEs and economic growth. Although the ARDL framework partially addresses this through lag structures, the results should be interpreted as indicative of associations rather than strict causality.

Third, the employment variable, measured as agricultural employment share, does not capture employment effects across the entire NTAE value chain, particularly in downstream sectors such as processing and logistics.

3.17 Justification of Methods

Non-Traditional Agricultural Exports (NTAEs) are measured using export values in monetary terms (US dollars), obtained from secondary data sources such. Employment is measured using indicators such as employment in the agricultural/export sector, depending on data availability, typically sourced from national labour statistics. The socio-ecological environment is proxied using an environmental indicator such as the ecological footprint, which captures the environmental impact of economic activities. These measurements are appropriate for time-series econometric analysis and ensure consistency with existing empirical studies, thereby supporting the reliability and validity of the results.

In conclusion, this chapter has outlined the methodological framework used in the study, including the research design, data sources, variable definitions, and estimation techniques. The ARDL model is adopted as the primary analytical tool, given its suitability for small samples and mixed integration orders. The next chapter presents the empirical results and their interpretation.

4 DATA ANALYSIS AND DISCUSSION OF RESULTS

4.1 Overview of Data and Analytical Approach

This chapter presents and discusses the empirical findings on the relationship between Ghana's Non-Traditional Agricultural Exports (NTAEs) and selected macroeconomic and environmental variables. The analysis is based on annual time-series data spanning 2000 to 2025, with particular emphasis on recent developments in 2024 and 2025 to enhance the policy relevance of the study.

Two complementary econometric techniques are employed: (i) the Autoregressive Distributed Lag (ARDL) model to capture both short-run and long-run dynamics under mixed orders of integration and (ii) Error Correction Model (ECM) to estimate contemporaneous relationships. Given the relatively small sample size, results are interpreted with caution, with attention to potential structural breaks and recent macroeconomic shifts.

4.2 Descriptive Statistics

The description of the descriptive statistics shown in Table 5 is a summary of the main component of the descriptive statistics of the study. They include the mean, the standard deviation, the number of observations and the range of the observations. The descriptive statistics are intended to provide a general overview of the extent of the data.

Table 5 Summary Statistics, using the observations 2000-2025

Variable	Mean	Median	Std. Dev.	Minimum	Maximum	Skewness	Ex. kurtosis
NTAE	0.30462	0.31000	0.18511	0.080000	0.71000	0.32547	-1.0095
Other NTE	1.8604	2.1050	0.89462	0.70000	3.4500	0.22080	-1.1589
Traditional Exp	32.733	29.045	16.094	9.1300	77.420	0.70938	0.38436
GDP	118.82	120.24	49.435	53.090	207.83	0.22217	-1.2789
Gov't Consumption	10.256	10.580	2.9772	5.3700	14.630	-0.17955	-1.1847
Imports	40.137	40.390	12.629	16.190	74.820	0.33131	0.75444
Investment	28.879	31.130	9.1081	11.540	40.440	-0.59899	-0.93098
Consumption	81.318	84.820	41.581	14.610	157.67	0.14634	-0.96021
Rural Agric Lab. Par. rate	57.058	58.000	9.3009	41.500	71.000	-0.18699	-1.2350
Agriculture Employment	48.229	45.050	10.597	34.570	64.000	0.28130	-1.4464
Agric workers wage	554.65	529.50	149.26	364.00	849.00	0.45308	-0.99947
Emp-to-pop. Ratio	67.365	67.400	1.4538	65.000	69.600	-0.087683	-1.2762
Inflation	16.157	13.670	8.6073	6.7000	39.210	1.2112	0.61472
Deforestation Rate	6.6269	4.0000	5.2804	0.50000	18.000	0.69086	-0.96528
Agric GH. gas Emissions	9.4281	9.3350	1.5318	6.9800	12.100	0.19256	-0.98036
Fertilizer use (kg/ha)	21.981	19.950	12.758	6.2000	50.100	0.47118	-0.95045
NTAE Production (Mt/ha)	11.873	11.750	2.6786	7.8000	16.000	0.065568	-1.3140
Agric Energy con. (Ktoe)	76.423	64.000	39.345	29.000	147.00	0.24352	-1.3977
EPI	40.237	42.050	7.1806	27.350	53.000	-0.49162	-0.74080

Source: Author's own design with results from Gretl

The summary statistics depicted in table 5 reveals important empirical insights into the behaviour of the key variables underpinning economic growth, employment dynamics, and environmental sustainability over the study period. Regarding economic growth, GDP records a mean value of 118.82 and a median of 120.24, indicating steady growth over time with limited influence from extreme values. However, the relatively high standard deviation of 49.44 and a wide range (minimum 53.09, maximum 207.83) suggest substantial growth fluctuations, reflecting structural changes and exposure to external economic shocks. Traditional exports show a high mean of 32.73 and a large dispersion ($SD = 16.09$), signalling heavy reliance on commodity-based exports and vulnerability to price volatility. By contrast, non-traditional agricultural exports (NTAE), with a mean of 0.30 and lower dispersion ($SD = 0.19$), exhibit more gradual and stable growth patterns, supporting their role in export diversification. Investment (mean = 28.88) and consumption (mean = 81.32) display notable variability, highlighting their significance as demand-side drivers of GDP.

Turning to employment-related variables, rural agricultural labour participation averages 57.06%, while agricultural employment averages 48.23%, both showing moderate dispersion, which indicates a gradual structural transition rather than abrupt labour shifts. Agricultural workers' wages exhibit relatively high variability ($SD = 149.26$) and a positively skewed distribution, suggesting uneven income distribution within the agricultural sector. The employment-to-population ratio, however, shows remarkable stability with a low standard deviation (1.45), implying that overall labour absorption remained structurally consistent despite economic fluctuations. These patterns suggest employment rigidity at the macro level, even as sectoral employment and wages evolve.

With respect to environmental factors, deforestation rates show considerable variability ($SD = 5.28$) and a strong positive skewness, indicating episodic spikes likely associated with agricultural expansion. Fertilizer use also varies substantially ($SD = 12.76$), suggesting uneven intensification of agricultural practices. Agricultural greenhouse gas emissions remain relatively stable ($SD = 1.53$), implying gradual environmental impacts rather than abrupt changes. Energy consumption in agriculture exhibits high dispersion ($SD = 39.35$), reflecting increased mechanisation. Finally, the Environmental Performance Index (EPI), with a mean of 40.24 and moderate variability, points to fluctuating environmental governance outcomes.

The summary statistics reveal meaningful variability across economic, employment, and environmental indicators, justifying the application of dynamic econometric techniques such as ARDL and ECM in subsequent sections of this Chapter.

4.3 Justification for Selecting the Variables Used

The selection of GDP, employment, and environmental variables in measuring the contribution of non-traditional agricultural exports (NTAEs) is empirically justified because NTAEs influence economic performance through multiple transmission channels that extend beyond trade revenues alone.

GDP captures the aggregate growth effect of NTAEs via export diversification, foreign exchange earnings, and value-chain expansion, while investment, consumption, imports, and government expenditure account for demand-side and macroeconomic adjustment mechanisms commonly highlighted in open-economy growth models. Employment-related variables such as agricultural employment, rural labour participation, wages, and the employment-to-population ratio are essential because NTAEs are labour-intensive and directly affect rural livelihoods, income distribution, and structural transformation in developing economies (Diao, McMillan, & Rodrik, 2019). Environmental indicators, including deforestation, fertilizer use, agricultural energy consumption, greenhouse gas emissions, and the Environmental Performance Index (EPI), are equally relevant because expansion in export-oriented agriculture inevitably interacts with land use, input intensification, and environmental sustainability, making it necessary to assess whether export-led growth is environmentally sustainable (FAO, 2022). The joint modeling of these variables is consistent with recent empirical studies that employ ARDL and ECM frameworks to examine trade-growth-environment-employment nexuses. Shahzad et al. (2021) and Khan et al. (2022) use ARDL-ECM models to link exports, GDP growth, and environmental outcomes, while Agboola and Bekun (2019) and Asongu et al. (2023) demonstrate the suitability of such models for capturing short- and long-run dynamics in developing-country contexts with mixed integration orders. Consequently, the chosen variables can broadly be viewed as theoretically grounded, empirically supported by recent literature, and methodologically appropriate for evaluating the multidimensional contribution of NTAEs to GDP, employment, and environmental sustainability.

4.4 Correlation Matrix

The correlation analysis examines the pairwise associations among NTAE, GDP, and the selected macroeconomic control variables such as traditional exports, imports, investment, government consumption, and household consumption in order to provide preliminary evidence on how these variables move together over time. By showing relationship strength and direction before modeling, the correlation matrix validates theoretical links and flags interdependencies or multicollinearity risks.

The correlation coefficients in **table 6** reveal strong multicollinearity among the explanatory variables, as several pairwise correlations are excessively high. Notably, NTAE is strongly correlated with other non-traditional exports ($r = 0.936$), traditional exports ($r = 0.897$), and GDP ($r = 0.967$), while GDP and consumption show an almost perfect correlation ($r = 0.989$). Additional high correlations, such as between traditional exports and imports ($r = 0.956$), further confirm close co-movement among regressors. However, this potential multicollinearity is methodologically addressed in the study through the adoption of ARDL and ECM approaches, which incorporate lag structures to distinguish between short-run dynamics and long-run equilibrium relationships, thereby mitigating multicollinearity concerns.

Table 6 Correlation coefficients, using the observations 2000 - 2025

Variables	NTAE	Other NTE	Trad. Exp	GDP	Gov't Exp.	Imports	Investment	Consumption
NTAE	1.0000	0.9357	0.8974	0.9674	0.7973	0.8136	0.5626	0.9414
Other NTE		1.0000	0.8408	0.9771	0.8876	0.8090	0.5435	0.9778
Traditional Exp			1.0000	0.8759	0.7107	0.9556	0.2451	0.8202
GDP				1.0000	0.8233	0.8019	0.5851	0.9887
Gov't Consumption					1.0000	0.7561	0.4882	0.8350
Imports						1.0000	0.1437	0.7574
Investment							1.0000	0.5942
H. Consumption								1.0000

Source: Author's own design with results from Gretl

4.5 Unit Root Test (ADF)

The Augmented Dickey-Fuller (ADF) unit root test was conducted to examine the stationarity properties of all the variables over the period 2000-2025. The test was performed with a constant (intercept) and automatic lag selection based on the Akaike Information Criterion (AIC), with a maximum of two lags, which is appropriate for annual data and aligns with ARDL requirements. The ADF unit root test is to guide the appropriate econometric modelling strategy. The null hypothesis of the ADF test states that a variable contains a unit root, implying non-stationarity. The results indicate that at levels, all the variables reported in the table are non-stationary at the 5 percent significance level.

Below is the ADF unit root test results;

Table 7 ADF test with intercept; lag length selected automatically using AIC (max lag = 2)

Variable	At Level			At 1st Difference				Order of Integration
	ADF Statistic	5% Critical Value	Decision	ADF Statistic	5% Critical Value	Decision	P-Value	
GDP	2.4368	-2.9862	Non-Stationary	-3.1644	-2.9919	Stationary	0.035	I(1)
NTAE	0.5350	-2.9981	Non-Stationary	-5.4025	-2.9981	Stationary	0.0002	I(1)
Other NTE	-0.0674	-2.9862	Non-Stationary	-4.9435	-2.9919	Stationary	0.0006	I(1)
Traditional Exports*	1.0133	-2.9862	Non-Stationary	-2.4303	-2.9919	Non-Stationary	0.1445	I(2)*
Imports	-0.2166	-2.9862	Non-Stationary	-4.3005	-2.9919	Stationary	0.0027	I(1)
Investment	-2.1007	-2.9862	Non-Stationary	-4.8935	-2.9919	Stationary	0.0007	I(1)
H. Consumption	0.8504	-2.9919	Non-Stationary	-6.1773	-2.9919	Stationary	0.0000	I(1)
Gov't Consumption	-1.8822	-2.9862	Non-Stationary	-6.3017	-2.9919	Stationary	0.0000	I(1)

Source: Author's own design with results from EViews

GDP records an ADF statistic of 2.4368, which is higher than the 5 percent critical value of -2.9862, leading to the failure to reject the null hypothesis. Similarly, NTAE (ADF = 0.5350), Other NTE (ADF = -0.0674), Traditional Exports (ADF = 1.0133), Imports (ADF = -0.2166), Investment (ADF = -2.1007), Household Consumption (ADF = 0.8504), and Government Consumption (ADF = -1.8822) are all non-stationary at levels. This indicates that these macroeconomic variables exhibit persistence and are influenced by long-run structural trends and shocks in the Ghanaian economy.

After first differencing, most of the variables become stationary at the 5 percent level. GDP attains stationarity with an ADF statistic of -3.1644, which is more negative than the critical value of -2.9919, with a corresponding p-value of 0.035. NTAE also becomes stationary after first differencing (ADF = -5.4025, p-value = 0.0002), indicating strong rejection of the unit root hypothesis. Other NTE, Imports, Investment, Household Consumption, and Government Consumption similarly record highly negative ADF statistics at first difference, all exceeding the 5 percent critical values, confirming stationarity and suggesting order of integration $I(1)$.

The null hypothesis of the ADF test posits that Traditional Exports contain a unit root. At level form, the reported ADF test statistic is 1.0133, with an associated p-value of 0.9954. This statistic is far less negative than the critical values at the 1 percent (-3.724), 5 percent (-2.986), and 10 percent (-2.633) significance levels. As a result, the null hypothesis cannot be rejected, indicating that Traditional Exports are non-stationary in levels.

Following first differencing, the ADF test statistic for Δ Traditional Exports is -2.430, with a p-value of 0.1445. This value remains less negative than the corresponding critical values at the 1 percent (-3.738), 5 percent (-2.992), and 10 percent (-2.636) levels. Consequently, the null hypothesis of a unit root cannot be rejected at first difference, implying that Traditional Exports remain non-stationary after first differencing.

However, after second differencing, the ADF test statistic becomes -5.7985, with a p-value of 0.0001. This statistic is more negative than all relevant critical values, leading to rejection of the null hypothesis. Thus, Traditional Exports are stationary only after second differencing, implying integration of order two, $I(2)$. Since the ARDL bounds testing approach requires variables to be at most $I(1)$, Traditional Exports are excluded from the long-run ARDL specification to ensure valid and consistent estimation.

4.6 ARDL Results for relationship between NTAEs and GDP (2000-2025)

Based on 25 effective observations after sample corrections, this section offers an empirical interpretation of the estimated ARDL (1,1,0,0,1,1,1) model that analyses the factors influencing GDP from 2001 to 2025. In order to ensure a parsimonious yet dynamically relevant representation of the underlying growth process, the model was chosen among 64 alternative specifications using the Akaike Information Criterion.

See the results from the ARDL model run for NTAEs and GDP below;

Table 8 ARDL Results for relationship between NTAEs and GDP (2000-2025)

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
GDP(-1)	0.565402	0.107250	5.271789	0.0002
NTAE	25.66777	8.535028	3.007345	0.0101
NTAE(-1)	17.37096	11.31534	1.535168	0.1487
OTHERNTE	2.770443	3.125845	0.886302	0.3916
IMPORTS	0.114775	0.103963	1.103992	0.2896
CONSUMPTION	0.480665	0.118659	4.050827	0.0014
CONSUMPTION(-1)	-0.164784	0.120369	-1.368990	0.1942
INVESTMENT	0.264732	0.109019	2.428318	0.0304
INVESTMENT(-1)	-0.218146	0.108963	-2.002024	0.0666
GOVTCON_CONSTANTLCU	-0.187024	0.449754	-0.415836	0.6843
GOVTCON_CONSTANTLCU (-1)	-0.410111	0.268162	-1.529340	0.1501
C	10.47970	4.674937	2.241678	0.0431
R-squared	0.999193	Mean dependent var	121.4448	
Adjusted R-squared	0.998511	S.D. dependent var	48.56427	
S.E. of regression	1.874159	Akaike info criterion	4.400271	
Sum squared resid	45.66213	Schwarz criterion	4.985331	
Log likelihood	-43.00338	Hannan-Quinn criter.	4.562541	
F-statistic	1463.824	Durbin-Watson stat	2.223378	
Prob(F-statistic)	0.000000			

Source: Author's own design with results from EViews

4.6.1 Interpretation

From Table 8, the lagged GDP coefficient is positive and highly significant (0.565, $p < 0.01$), indicating strong growth persistence. This shows that current output is largely driven by past performance, validating the dynamic ARDL framework and suggesting that economic shocks have lasting effects that extend into subsequent periods. NTAEs exhibit a positive and statistically significant contemporaneous impact on GDP (25.67, $p = 0.010$), providing strong empirical support for the export-led growth hypothesis and highlighting the role of agricultural export diversification in economic growth. The lagged NTAE coefficient is positive but statistically insignificant, indicating that NTAE-driven growth effects are largely short-run in nature, transmitted through income, employment, and foreign exchange generation. Other non-traditional exports record positive yet insignificant coefficients, suggesting their growth effects are indirect or operate through complementary macroeconomic transmission channels rather than direct output expansion.

Imports record a positive but statistically insignificant short-run coefficient (0.115, $p = 0.29$), indicating no direct growth effect. This likely reflects an import structure dominated by consumption goods rather than capital or intermediate inputs, which increases demand leakage without expanding productive capacity, thereby limiting their immediate contribution to GDP growth.

Household consumption emerges as a key driver of economic growth, with a positive and highly significant coefficient (0.481, $p < 0.01$). This highlights the central role of private demand in sustaining economic activity and supports the view that growth during the period was largely demand-driven. Investment also has a positive and statistically significant contemporaneous effect on GDP (0.265, $p = 0.030$), confirming the importance of capital formation in expanding productive capacity. However, the negative lagged investment coefficient reflects adjustment costs and gestation lags associated with investment projects.

Government expenditure carries negative coefficients and remains statistically insignificant in both current and lagged forms. This suggests that public expenditure during the period did not directly contribute to growth, possibly due to its predominance in recurrent spending such as wages and administrative costs rather than productivity-enhancing capital investment. Such spending may also crowd out private investment by absorbing limited fiscal and financial resources. All things considered, the model exhibits good explanatory power and honest diagnostics, suggesting that NTAEs, household spending, and private investment are the main drivers of economic growth, with imports and government consumption having a minimal direct impact.

4.6.2 Bounds Test

The ARDL bounds testing approach was used to assess whether a long-run equilibrium relationship exists between GDP and NTAE, other NTEs, imports, investment, government consumption, and household consumption for 2000-2025, under a restricted constant and no-trend specification, with the null hypothesis assuming no long-run relationship.

As observed in Table 9 below, the computed F-statistic from the bounds test is 40.99, which substantially exceeds the upper bound critical values at the 10%, 5%, and 1% significance levels for both asymptotic and finite-sample distributions. As a result, the null hypothesis of no cointegration is decisively rejected, providing strong empirical evidence of a stable long-run relationship between GDP and the included explanatory variables. This finding implies that changes in exports particularly non-traditional agricultural and other non-traditional exports alongside investment and consumption are systematically linked to long-term economic performance.

Further confirmation of cointegration is provided by the error-correction term derived from the conditional error-correction regression. The coefficient of the lagged GDP term is negative and highly significant (-0.646, $p < 0.01$), indicating that approximately 65% of short-run deviations from long-run equilibrium are corrected within one period. This rapid speed of adjustment suggests that shocks to exports, investment, or consumption have only temporary effects, with GDP converging quickly back to its long-run path. Overall, the bounds test results validate the use of the ARDL-ECM framework for analyzing both long-run and short-run dynamics in the study.

Table 9 ARDL Long Run Form and Bonds Test

Conditional Error Correction Regression				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	10.47970	4.674937	2.241678	0.0431
GDP(-1)*	-0.434598	0.107250	-4.052179	0.0014
NTAE(-1)	43.03873	14.56278	2.955392	0.0112
OTHERNTE**	2.770443	3.125845	0.886302	0.3916
IMPORTS**	0.114775	0.103963	1.103992	0.2896
CONSUMPTION(-1)	0.315882	0.107177	2.947280	0.0113
INVESTMENT(-1)	0.046586	0.101307	0.459854	0.6532
GOVTCON_CONSTANT...	-0.597135	0.548700	-1.088272	0.2962
D(NTAE)	25.66777	8.535028	3.007345	0.0101
D(CONSUMPTION)	0.480665	0.118659	4.050827	0.0014
D(INVESTMENT)	0.264732	0.109019	2.428318	0.0304
D(GOVTCON CONSTA...	-0.187024	0.449754	-0.415836	0.6843
* p-value incompatible with t-Bounds distribution.				
** Variable interpreted as $Z = Z(-1) + D(Z)$.				
Levels Equation Case 2: Restricted Constant and No Trend				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
NTAE	99.03110	28.00987	3.535578	0.0037
OTHERNTE	6.374723	7.026523	0.907237	0.3808
IMPORTS	0.264094	0.226314	1.166937	0.2642
CONSUMPTION	0.726836	0.145855	4.983294	0.0003
INVESTMENT	0.107194	0.237274	0.451773	0.6589
GOVTCON_CONSTANT...	-1.373993	1.146960	-1.197943	0.2523
C	24.11355	7.901468	3.051781	0.0093
EC = GDP - (99.0311*NTAE + 6.3747*OTHERNTE + 0.2641*IMPORTS + 0.7268*CONSUMPTION + 0.1072*INVESTMENT -1.3740 *GOVTCON_CONSTANTLCU_ + 24.1136)				
F-Bounds Test				
Null Hypothesis: No levels relationship				
Test Statistic	Value	Signif.	I(0)	I(1)
Asymptotic: n=1000				
F-statistic	6.759920	10%	1.99	2.94
k	6	5%	2.27	3.28
		2.5%	2.55	3.61
		1%	2.88	3.99
Finite Sample: n=30				
Actual Sample Size	25	10%	2.334	3.515
		5%	2.794	4.148
		1%	3.976	5.691

Source: Author's own design with results from EViews 12

4.6.3 Results ECM Regression

The ECM results in table 10 indicates The estimated error-correction term, CointEq(-1), is negative and highly statistically significant (-0.435, $p < 0.01$), confirming the existence of a stable long-run equilibrium relationship between GDP and its explanatory variables. The magnitude of this coefficient indicates that approximately 43 percent of short-run deviations from long-run equilibrium are corrected within one year, reflecting a relatively fast adjustment process following economic shocks.

In the short run, changes in non-traditional agricultural exports (Δ NTAE) exert a strong positive and statistically significant effect on GDP (25.67, $p < 0.01$), highlighting the immediate growth benefits of export diversification. Household consumption also shows a positive and highly significant impact (0.481, $p < 0.01$), underscoring the importance of private demand as a key driver of economic growth. Similarly, investment has a positive and statistically significant short-run effect on output (0.265, $p < 0.01$), confirming the role of capital formation in enhancing productive capacity.

See table 10 for the ECM Regression results conducted on the model below;

Table 10 ADRL Error Correction Regression

ECM Regression Case 2: Restricted Constant and No Trend				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(NTAE)	25.66777	4.519499	5.679340	0.0001
D(CONSUMPTION)	0.480665	0.053968	8.906412	0.0000
D(INVESTMENT)	0.264732	0.053601	4.938920	0.0003
D(GOVTCON CONSTA...)	-0.187024	0.166473	-1.123446	0.2816
CointEq(-1)*	-0.434598	0.047646	-9.121349	0.0000
R-squared	0.835495	Mean dependent var		6.189600
Adjusted R-squared	0.802594	S.D. dependent var		3.400817
S.E. of regression	1.510995	Akaike info criterion		3.840271
Sum squared resid	45.66213	Schwarz criterion		4.084046
Log likelihood	-43.00338	Hannan-Quinn criter.		3.907883
Durbin-Watson stat	2.223378			

* p-value incompatible with t-Bounds distribution.

F-Bounds Test		Null Hypothesis: No levels relationship		
Test Statistic	Value	Signif.	I(0)	I(1)
F-statistic	6.759920	10%	1.99	2.94
k	6	5%	2.27	3.28
		2.5%	2.55	3.61
		1%	2.88	3.99

Source: Author's own design with results from EViews 12

On the other hand, changes in government consumption expenditure have a negative but statistically insignificant effect (-0.187, $p = 0.28$), indicating that public spending did not directly boost economic development during that time, maybe because it was mostly recurrent.

With an adjusted R-squared of 0.803, the model demonstrates good explanatory power, meaning that the variables included account for more than 80% of short-run fluctuations in GDP. The ARDL-ECM framework is validated and cointegration is further confirmed by the limits test F-statistic of 6.76, which surpasses the upper critical bounds at all conventional significance levels.

To conclude, GDP and non-traditional agricultural exports (NTAE) have a robust and consistent long-term association, according to the ARDL data. The significant error-correction term shows quick adjustment to equilibrium, confirming the export-led growth function of NTAEs during the research period. NTAE has a positive and statistically significant short-run impact on GDP.

4.7 Results for relationship between NTAEs and Employment (2000-2025)

The employment-to-population ratio in Ghana is experimentally examined in this session using an Autoregressive Distributed Lag (ARDL) framework from 2001 to 2025. Due to the variables' heterogeneous integration orders and the sample's small size of 25 observations, the ARDL technique is suitable and produces reliable short-run estimates. The presence of dynamic labor-market adjustment is confirmed by the identification of an ARDL(1,0,0,1,0,1) specification through model selection based on the Akaike Information Criterion.

Below is the result of the ARDL output indicating the relationship between NTAE and Employment in Ghana;

Table 11 ARDL Results for relationship between NTAEs and Employment (2000-2025)

Selected Model: ARDL(1, 0, 0, 1, 0, 1)

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
EMPLYMENTTOPOPULATIONRATIO(-1)	1.026892	0.040291	25.48679	0.0000
NTAE	1.577901	0.697262	2.262994	0.0379
ANNUALAGRICWORKERSWAGEUSD01	-0.013650	0.004534	-3.010645	0.0083
INFLATION	0.003400	0.004569	0.743974	0.4677
INFLATION(-1)	0.011129	0.005368	2.073270	0.0547
RURALAGRICLABOURPART_RATE	-0.124756	0.065274	-1.911272	0.0741
GDP	-0.021308	0.014212	-1.499275	0.1533
GDP(-1)	0.041319	0.011996	3.444265	0.0033
C	10.01879	5.031335	1.991278	0.0638
R-squared	0.992940	Mean dependent var		67.28400
Adjusted R-squared	0.989410	S.D. dependent var		1.422053
S.E. of regression	0.146342	Akaike info criterion		-0.732034
Sum squared resid	0.342654	Schwarz criterion		-0.293239
Log likelihood	18.15042	Hannan-Quinn criter.		-0.610331
F-statistic	281.2808	Durbin-Watson stat		2.450407
Prob(F-statistic)	0.000000			

Source: Author's own design with results from Gretl

4.7.1 Interpretation

The lagged employment-to-population ratio has a coefficient of 1.0269 and is highly significant ($p < 0.01$), indicating strong persistence in employment outcomes in Ghana. NTAE exerts a positive and statistically significant effect ($\beta = 1.5779$; $p = 0.038$), implying that improvements in this variable enhance employment absorption. Conversely, agricultural workers' wages negatively affect employment ($\beta = -0.0137$; $p = 0.008$), reflecting structural rigidities and low productivity in Ghana's agricultural sector.

Inflation is insignificant contemporaneously, but its lag has a marginal positive effect ($\beta=0.0111$; $p=0.055$), suggesting delayed macroeconomic transmission. Rural agricultural labour participation reduces employment intensity ($\beta=-0.1248$; $p=0.074$), consistent with disguised unemployment in rural Ghana. GDP has a positive and significant lagged effect ($\beta=0.0413$; $p=0.003$), indicating that economic growth translates into employment gains with a time lag.

4.7.2 Unit Root Test (ADF)

Table 12 reports the Augmented Dickey-Fuller (ADF) unit root test results for the employment-to-population ratio in Ghana. The ADF unit root test results show that all variables are non-stationary in levels at the 5% significance level, as their test statistics are less negative than the respective critical values. For instance, the employment-to-population ratio has an ADF statistic of -2.2647, while GDP records 2.4368, both failing to reject the null hypothesis of a unit root. However, after first differencing, all variables become stationary, with highly negative ADF statistics such as -13.1195 for employment-to-population ratio, -5.4025 for NTAE, and -3.1644 for GDP. The corresponding p-values, all below 5%, confirm significance. Thus, all series are integrated of order one, $I(1)$, justifying the use of the ARDL modelling framework. Below are the results of the Unit root test;

Table 12 ADF test with intercept; lag length selected automatically using AIC (max lag = 2)

Variable	At Level			At 1st Difference				Order of Integration
	ADF Statistic	5% Critical Value	Decision	ADF Statistic	5% Critical Value	Decision	P-Value	
Emp.-to-Pop. Ratio	-2.2647	-2.9919	Non-Stationary	-13.1195	-2.9919	Stationary	0.034	$I(1)$
NTAE	0.5349	-2.9981	Non-Stationary	-5.4025	-2.9981	Stationary	0.0002	$I(1)$
Agric. Workers' Wage	4.8030	-2.9981	Non-Stationary	-2.3252	-2.9981	Stationary	0.031	$I(1)$
Inflation	-2.8675	-2.9862	Non-Stationary	-2.8675	-2.9862	Stationary	0.0087	$I(1)$
Rural Agric. Lab. Part. Rate	2.0377	-2.9862	Non-Stationary	-2.0377	-2.9862	Stationary	0.0000	$I(1)$
GDP	2.4368	-2.9862	Non-Stationary	-3.1644	-2.9862	Stationary	0.023	$I(1)$

Source: Author's own design with results from EViews 12

4.8 Results Output on NTAE and the Environment Relationship

This section presents the empirical results from the analysis of the relationships among Non-Traditional Agricultural Exports (NTAE) and ecological footprints.

4.8.1 ARDL (1,1) NTAE and Ecological Footprint (Greenhouse Emissions)

Table 13 NTAE and Ecological Footprint

<i>Variables</i>	<i>Coefficient</i>	<i>Std. Error</i>	<i>t-ratio</i>	<i>p-value</i>	
const	1.75069	0.0811149	21.58	<0.0001	***
lfert	0.180489	0.0262915	6.865	<0.0001	***
ldefor	0.0407316	0.0183807	2.216	0.0379	**
d_lntp	-4.00043	1.40001	-2.857	0.0094	***
Mean dependent var	2.233186	S.D. dependent var		0.166211	
Sum squared resid	0.045144	S.E. of regression		0.046365	
R-squared	0.931912	Adjusted R-squared		0.922186	
F (3, 21)	95.80879	P-value(F)		2.05e-12	
Log-likelihood	43.48620	Akaike criterion		-78.97240	
Schwarz criterion	-74.09690	Hannan-Quinn		-77.62015	
rho	-0.262030	Durbin-Watson		2.520016	

Source: Author's own design with results from Gretl

4.8.2 Interpretation

As observed from the table above, the regression is highly significant ($F = 95.89$, $p < 0.001$) and explains about 92 % of the variation in emissions ($R^2 = 0.922$), indicating strong model fit. The constant term (1.75, $p < 0.001$) reflects a substantial baseline emission level. Fertilizer use ($lfert = 0.18$, $p < 0.001$) and deforestation ($ldefor = 0.04$, $p = 0.038$) both exert positive and statistically significant effects, confirming that intensive agricultural practices and land-use change increase emissions. Conversely, the change in NTAEs ($d_lntp = -4.00$, $p = 0.009$) has a significant negative effect, implying that NTAE expansion contributes to emission reduction likely through diversification, improved efficiency, and reduced pressure on forest resources. The Durbin-Watson = 2.52 suggests no autocorrelation, supporting model reliability.

Overall, the results align with the environmental sustainability hypothesis: while conventional intensification worsens ecological outcomes, export diversification mitigates them. This model effectively captures the short-run interaction between NTAE growth and environmental performance, demonstrating that Ghana's pursuit of non-traditional exports can promote sustainable agricultural development when accompanied by responsible land and input management.

4.8.3 Diagnostic Tests

Table 14 Heteroskedasticity (White's test)

	coefficient	std. error	t-ratio	p-value
const	0.0398108	0.0644828	0.6174	0.5462
lfert	-0.0302136	0.0327723	-0.9219	0.3712
ldefor	0.0103410	0.0134003	0.7717	0.4523
d_lntp	0.282681	1.81842	0.1555	0.8785
sq_lfert	0.00427632	0.00422044	1.013	0.3270
X2_X3	-0.00147923	0.00473075	-0.3127	0.7588
X2_X4	0.125897	0.513098	0.2454	0.8095
sq_ldefor	-0.000689110	0.00191563	-0.3597	0.7241
X3_X4	-0.0757735	0.267215	-0.2836	0.7806
sq_d_lntp	-10.7673	14.7280	-0.7311	0.4760

Unadjusted R-squared = 0.419933

Test statistic: $TR^2 = 10.498334$,
with p-value = $P(\text{Chi-square}(9) > 10.498334) = 0.311667$

Source: results from Gretl

The output above presents White's test for heteroskedasticity, which checks whether the variance of residuals is constant across observations. The test statistic ($TR^2 = 10.50$, $p = 0.312$) is well above the 0.05 threshold, meaning the null hypothesis of homoskedasticity cannot be rejected. This indicates that the model's residuals have a constant variance, and there is no evidence of heteroskedasticity. All explanatory and squared interaction terms show insignificant p-values, reinforcing that none systematically affect error variance.

4.8.4 Autocorrelation Breusch-Godfrey test

The result shows the Breusch-Godfrey test for first-order autocorrelation in the residuals (uhat) of the regression model. The variables included are const, lfert (fertilizer use), ldefor (deforestation), d_lntp (change in NTAEs Production), and uhat_1 (lagged residual). All coefficients are statistically insignificant ($p > 0.05$), meaning none of these variables explain the residual pattern. The test statistics LMF = 1.83 ($p = 0.191$), $TR^2 = 2.10$ ($p = 0.147$), and Ljung-Box $Q' = 1.92$ ($p = 0.166$) are above the 0.05 threshold, confirming no first-order autocorrelation. Thus, the residuals are independent, validating the model's efficiency and reliability for inference.

To sum it all, the ARDL model evaluates the sustainability of NTAEs by linking export activities to ecological footprints, critical for achieving SDG 12 (Responsible Consumption and Production). It also quantifies immediate economic benefits, informing policies to enhance NTAE contributions to GDP (SDG 8: Decent Work and Economic Growth).

By integrating these approaches, the study bridges gaps in existing literature, providing actionable insights for Ghana's policymakers to balance economic growth with socio-ecological stewardship. There was a relationship between NTAE production and Socio-Ecological factors such as deforestation, greenhouse emissions (carbon dioxide emissions), forest cover lost, fertilizer usage, etc. Since NTAE production has a relationship with socio-ecological factors, it also implies that the export of NTAE also has a relationship with socio-ecological factors such as carbon dioxide emissions and deforestation rate.

4.9 Key Findings

The empirical results in Chapter 4 provide robust evidence on the multidimensional role of Non-Traditional Agricultural Exports (NTAEs) in Ghana's economy between 2000 and 2025. First, the growth analysis confirms a strong export-led growth effect of NTAEs. The ARDL results show that NTAEs exert a positive and statistically significant short-run impact on GDP, with a coefficient of 25.67 ($p < 0.01$), indicating that a one-unit increase in NTAE value significantly expands national output. The bounds test further confirms a stable long-run cointegrating relationship, while the error-correction term (-0.646 ; $p < 0.01$) indicates that about 65% of short-run disequilibria in GDP are corrected within one year. Empirically, this supports the export-led growth hypothesis and highlights NTAEs as a key mechanism for diversifying Ghana's export base away from volatile traditional commodities.

More so, the employment results reveal that NTAEs influence labour outcomes through structural transformation rather than pure job expansion in primary agriculture. Although NTAE growth is associated with a positive and significant effect on the employment-to-population ratio ($\beta = 1.58$; $p = 0.038$), agricultural employment shares declined from about 43% in 2015 to 34% in 2025. This suggests rising productivity, mechanisation, and a shift toward higher-value activities such as processing, logistics, and export services. GDP's lagged effect on employment ($\beta = 0.041$; $p = 0.003$) further indicates that employment responds to economic growth with a delay, reinforcing the dynamic nature of labour absorption.

Third, the environmental analysis highlights a trade-off between economic gains and ecological sustainability. While fertilizer use ($\beta = 0.18$; $p < 0.001$) and deforestation ($\beta = 0.04$; $p < 0.05$) significantly increase agricultural greenhouse gas emissions, changes in NTAEs exhibit a negative and significant effect on emissions ($\beta = -4.00$; $p < 0.01$). This suggests that export diversification toward NTAEs can mitigate environmental pressure when linked to efficiency gains and improved practices.

In all, the results show that NTAEs enhance GDP and employment quality, but sustainable policy frameworks are essential to manage environmental risks.

5 CONCLUSION AND RECOMMENDATION

This chapter presents the summary of the study, key conclusions drawn from the empirical findings, and policy recommendations. The study examined the socio-economic and environmental impacts of Non-Traditional Agricultural Exports (NTAEs) on Ghana's economy using time-series data from 2000 to 2025. The analysis focused on three core dimensions: economic growth (GDP), employment, and the socio-ecological environment.

5.1 Summary of Key Findings

5.1.1 NTAEs and Economic Growth

The empirical analysis produced several important findings:

The findings indicate a strong, positive, and statistically significant relationship between Non-Traditional Agricultural Exports (NTAEs) and economic growth in Ghana, thereby confirming the export-led growth hypothesis. This suggests that the expansion of NTAE exports contributes substantially to national output and plays a critical role in enhancing Ghana's economic performance. In addition, export diversification within the NTAE sector supports increased foreign exchange earnings and strengthens macroeconomic stability. The results further suggest that value addition within NTAEs enhances Ghana's integration into global value chains, thereby improving competitiveness in international markets. Moreover, recent macroeconomic improvements, including declining inflation and rising GDP growth, appear to reinforce the enabling environment for export expansion and sectoral growth.

5.1.2 NTAEs and Employment Dynamics

The relationship between NTAEs and employment is found to be complex and largely structural rather than purely quantitative. The findings suggest a decline in the share of agricultural employment despite increasing NTAE revenues, indicating a shift in the composition of labour within the sector. This reflects ongoing structural transformation, where labour is increasingly moving from traditional primary agricultural activities to higher-value segments such as agro-processing, logistics, and export-related services. This transition highlights a composition effect, whereby improvements in employment quality and productivity occur even as the proportion of labour engaged in primary agriculture declines. Furthermore, the results suggest that mechanization and commercialization within the NTAE sector reduce reliance on traditional farm labour while promoting more efficient and value-driven employment opportunities across the agricultural value chain.

5.1.3 NTAEs and the Socio-Ecological Environment

The findings reveal a significant relationship between the expansion of NTAEs and environmental degradation, highlighting important sustainability concerns. Increased NTAE activities are associated with rising deforestation rates and higher levels of carbon dioxide (CO₂) emissions, suggesting that export-driven agricultural growth exerts pressure on natural ecosystems.

In addition, the ARDL model confirms the existence of a long-run relationship between export growth and the ecological footprint, indicating that the environmental impact of NTAEs persists over time. These results underscore a critical trade-off between the economic benefits derived from NTAE expansion and the environmental costs associated with such growth. Consequently, while the sector contributes positively to economic development, it also raises significant challenges for environmental sustainability.

5.1.4 Conclusions

Based on the study's findings, Non-Traditional Agricultural Exports (NTAEs) serve as a key driver of Ghana's economic growth, playing a crucial role in export diversification and enhancing macroeconomic stability. The evidence further suggests that NTAEs promote structural transformation within the economy rather than directly expanding agricultural employment, as labour increasingly shifts toward higher-value segments such as agro-processing and export-related services. However, the expansion of NTAEs is also associated with environmental costs, as increased export-oriented agricultural activities contribute to ecological degradation in the absence of effective regulatory frameworks. In addition, the findings indicate that macroeconomic stability, characterized by lower inflation and sustained GDP growth, creates a conducive environment for the growth and performance of the NTAE sector.

5.2 Policy Recommendations

Ghana's Non-Traditional Agricultural Exports (NTAEs) have emerged as a vital driver of economic diversification, rural employment, and foreign exchange earnings. However, the econometric evidence from this study reveals that while NTAEs positively influence GDP and employment, they also pose environmental and structural challenges. To ensure long-term sustainability, policy interventions must integrate economic, social, and ecological dimensions.

- **The Feed Ghana initiative:** - The ARDL results presented by this study suggest that recent development strategies of the Government of Ghana such as the feed Ghana initiative which emphasizes increased agricultural productivity, value addition, and food security, is strongly supported by the positive and significant roles of non-traditional agricultural exports (NTAE) and investment in the ARDL results. These findings imply that strengthening agro-based production systems and linking them to export markets can enhance GDP both in the short and long run.
- **Grow for Industry:** - Similarly, the Grow for Industry policy aligns with the significant contribution of exports and investment, as it seeks to promote agro-processing and supply raw materials to domestic industries, thereby deepening industrial value chains and reducing import dependence. This strategy well aligns with the structural drivers of growth identified in the study, particularly export diversification, private investment, etc.

- **The 24-Hour Economy policy:** - This is also consistent with the empirical importance of household consumption, which appears as a strong and significant driver of growth. By extending productive and commercial activity beyond traditional working hours, this policy has the potential to stimulate private or domestic demand, employment, and production, reinforcing demand-led growth dynamics. In contrast, the insignificance of government consumption in both the ARDL and ECM results suggests that growth is more effectively driven by productivity-enhancing and private-sector-oriented policies rather than expansions in recurrent public expenditure.
- **Strengthening Sustainable Agricultural Practices:** - The government should make Environmental Impact Assessments (EIAs) mandatory for all export-oriented farming activities. This aligns with the Environmental Protection Agency's (EPA) mandate under the Environmental Protection Act (Act 490). EIAs will help mitigate deforestation, soil degradation, and water pollution associated with intensive export farming. Farmers should be incentivized to adopt climate-smart agriculture, organic farming, and agroforestry systems through targeted subsidies and technical support under the Planting for Food and Jobs (PFJ) and Planting for Export and Rural Development (PERD) programs. These practices will enhance productivity while preserving ecological integrity.
- **Promoting Value Addition and Industrialization:** - To maximize export earnings and employment, Ghana must shift from raw commodity exports to processed agricultural goods. The National Export Development Strategy (NEDS 2020-2029) already emphasizes value addition and market diversification. This should be reinforced through investment in agro-processing facilities, cold-chain logistics, and quality certification systems. Public-private partnerships can accelerate industrialization, reduce post-harvest losses, and create high-quality jobs. Export incentives should prioritize firms engaged in processing and packaging rather than raw commodity trading.
- **Enhancing Employment through Value Chain Development:** - Mechanization and export-driven modernization risk displacing rural labour. To counter this, the government should expand agribusiness training programs under the Ghana Jobs and Skills Project (2021-2026), focusing on youth and women. Support for small and medium-sized enterprises (SMEs) in processing, packaging, and logistics will foster inclusive participation across the NTAE value chain. These interventions will bridge the employment gap and sustain rural livelihoods.
- **Improving Access to Agricultural Finance:** - Smallholder farmers remain constrained by limited access to credit. The government, through the Ghana EXIM Bank, Agricultural Development Bank (ADB), and the proposed Women Development Bank, should establish low-interest export financing schemes and credit guarantees for NTAE producers. Financial inclusion will enable farmers to meet international quality standards and expand production sustainably.

- **Establishing a National NTAE Sustainability Index:** - A National NTAE Sustainability Index should be developed to monitor economic performance, employment impact, and environmental footprint. This index will serve as a policy evaluation tool, guiding resource allocation and ensuring evidence-based decision-making. It should be managed collaboratively by the Ministry of Trade and Industry (MoTI), Ministry of Food and Agriculture (MoFA), and the EPA.
- **Aligning Export Policies with the SDGs:** - Export strategies must align with SDG 8 (Decent Work and Economic Growth) and SDG 12 (Responsible Consumption and Production). Ghana should promote environmentally certified exports to enhance competitiveness in global markets. Certification schemes such as Fairtrade and Rainforest Alliance can be integrated into NTAE value chains to ensure sustainable production.
- **Strengthening Institutional Coordination and Land-Use Governance:** - Effective implementation requires stronger coordination among MoTI, MoFA, and EPA. The government should enforce sustainable land-use policies under the National Land Policy (1999) and the Forestry Commission's REDD+ Strategy to curb deforestation linked to export expansion. Institutional synergy will ensure coherent policy execution and long-term environmental sustainability.

5.5 Contribution to Knowledge

This study contributes to existing literature by offering a multi-dimensional analysis of Non-Traditional Agricultural Exports (NTAEs) in Ghana, with particular emphasis on their economic, employment, and environmental dimensions. Unlike previous studies that often focus on a single aspect of export performance, this research integrates these three dimensions to provide a more holistic understanding of the sector's development and its broader implications for sustainable growth.

In addition, the study advances methodological application in the Ghanaian context by employing Autoregressive Distributed Lag (ARDL) and regression models using time-series data spanning 2000 to 2025. This enhances empirical rigor in examining both short-run and long-run relationships between NTAEs, economic growth, employment, and environmental outcomes, thereby strengthening the robustness of findings within the field of agricultural export studies.

Furthermore, the research bridges a critical gap in the literature between export-led growth and environmental sustainability. While existing studies often emphasize the economic benefits of export expansion, this study highlights the accompanying environmental costs and trade-offs, thereby contributing to a more balanced discourse on sustainable agricultural development in emerging economies such as Ghana.

5.4 Limitations of the Study

Despite the contributions of this study, several limitations should be acknowledged. First, the relatively small sample size restricts the extent to which the findings can be generalized beyond the specific context of Ghana's Non-Traditional Agricultural Export (NTAE) sector. As a result, caution should be exercised when applying the conclusions to other countries or broader agricultural export contexts.

Second, the employment data used in the analysis does not fully capture all forms of job creation within the value chain. In particular, indirect and informal employment generated through supporting activities such as logistics, packaging, and small-scale processing may be underrepresented, thereby potentially understating the true employment impact of NTAEs.

Finally, the study may be affected by possible endogeneity between key variables, such as export performance, employment, and environmental outcomes. This implies that some of the observed relationships may be bidirectional, which could introduce bias in the estimated results. Consequently, the findings should be interpreted with caution, and future studies may consider more advanced econometric techniques to address this issue.

5.5 Suggestions for Future Research

Future studies on Non-Traditional Agricultural Exports (NTAEs) could benefit from the use of micro-level data, such as firm-level or household-level datasets, to provide a more detailed understanding of how export activities influence income distribution, productivity, and employment at the grassroots level. This would help to complement the macro-level insights provided in this study.

In addition, further research could explore regional disparities within Ghana to determine whether the impacts of NTAEs differ across ecological zones or administrative regions. Such an approach would help identify location-specific challenges and opportunities, thereby informing more targeted policy interventions.

Future studies may also focus on specific NTAE commodities such as cashew, shea, and horticultural products. A product-specific analysis would allow for a deeper examination of value chain dynamics, competitiveness, and sustainability issues within individual export subsectors.

Finally, there is scope for incorporating a broader range of environmental indicators beyond CO₂ emissions and deforestation. Variables such as soil degradation, water usage, biodiversity loss, and agrochemical pollution could provide a more comprehensive assessment of the environmental impacts of NTAE expansion.

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8 Attachments

9 Abbreviations

AfDB	African Development Bank
ADF	Augmented Dickey Fuller (Unit Root Test)
ARDL	Autoregressive Distributed Lag
ASCI	Africa Sustainable Commodities Initiative
BOG	Bank of Ghana
CoP	Conference of the Parties
CSA	Climate-Smart Agriculture
ECM	Error Correction Model
ECOWAS	Economic Community of West African States
EKC	Environmental Kuznets Curve
ELG	Export-Led Growth
EPA	Environmental Protection Authority, Ghana
EPI	Environmental Performance Index
GEPA	Ghana Export Promotion Authority
GVCs	Global Value Chains
EU	European Union
FAO	Food and Agriculture Organization
FDI	Foreign Direct Investment
GDP	Gross Domestic Product
GSS	Ghana Statistical Service
ILO	International Labour Organization
ILOSTAT	International Labour Organization Statistical Database
IMF	International Monetary Fund
Ktoe	Kilotonnes of Oil Equivalent
MOFA	Ministry of Food and Agriculture (Ghana)
MOTI	Ministry of Trade and Industry (Ghana)
Kha	Thousand hectares
Mt	Metric tonnes
NEDS	National Export Development Strategy
NTAEs	Non-Traditional Agricultural Exports
NTEs	Non-Traditional Exports
OECD	Organisation for Economic Co-operation and Development
PERD	Planting for Export and Rural Development
PFJ	Planting for Food and Jobs
PPP	Public-Private Partnership
REDD+	Reducing Emissions from Deforestation and Forest Degradation
SPS	Sanitary and Phytosanitary Standards
SDGs	Sustainable Development Goals
SMEs	Small and Medium-Sized Enterprises
UNCTAD	United Nations Conference on Trade and Development
WDI	World Development Indicators (World Bank)
WTO	World Trade Organization